

Microeconomics — End-Term Notes

IIMA PGP1 • Prof. Jeevant Rampal • open-book, open-notes

Built for the 2025-style paper: **12 descriptive questions, 10 marks each, 120 minutes, no negative marking, calculators allowed.** Everything is written in full sentences, so a section you have never read still makes sense on first pass. Factor markets are excluded (off syllabus). Numbers have been checked by computer.

How to use this in the hall

No negative marking means never leave a blank. Even a correct definition plus the right first-order condition earns partial credit. His own instruction on the 2025 paper says it explicitly: *“if the mathematics or calculations involved are beyond your understanding, please provide brief, clear and precise non-mathematical explanations to maximise partial credit.”* So if you cannot finish the algebra, write the economics.

Answers are graded on precision, not length. His rubric says that more than about 100 words on a short question is usually misdirected effort. Lead with the answer, then give the reason, then the check.

The four shapes of question he sets.

1. *A model you have seen, with the numbers changed.* Identify the model, write its first-order condition, solve, and sanity-check (Sections 1–24 here).
2. *A model you have not seen, defined inside the paper.* He gives you the payoff function; your job is to apply a standard method to it. The recipe is in §0.1.
3. *A verbal question about a real situation or an article.* Name the concept, say who bears the effect, and say why the market fails. Templates in §25.
4. *A ranking or classification question.* Rank pricing schemes, or sort situations into adverse selection versus moral hazard. Both are in §25.

0.1 Attacking a setup you have never seen

This is where the marks are won, and the 2025 securities auction (§8) is the clearest example. The method never changes.

1. **Write down the payoff function exactly as given.** Do not paraphrase it. In the 2025 debt auction the paper hands you $v - d + d^2/4v$; everything follows from that single line.
2. **Ask which standard question is being asked:** is this a dominance question, a best-response question, a backward-induction question, or an indifference condition? The verbs give it away — “dominant” means compare against every opponent action; “equilibrium” means mutual best responses; “efficient” means check whether the right person wins.
3. **Import the argument from the cash version.** A second-price auction of *anything* is solved by the rule “bid the amount that makes your winning payoff exactly zero,” because that is what makes over- and under-bidding both harmless. A first-price auction of anything is solved by maximising (probability of winning) $\times$ (payoff if you win).
4. **Solve the smallest case.** If N players are too many, do $N = 1, 2, 3$ and state the pattern. He gives generous credit for a correct induction.
5. **Sanity-check and say so in one clause.** “At the optimum all three marginal magnitudes equal 4000, as the rule requires.” One clause like this signals to the grader that you understood rather than pattern-matched.

0.2 Where to look, by keyword

If the question says...	Go to
dominant, dominated, maximin, best response, mixed	§1
integer / whole-number prices or bids, "choose a number between"	§2
moves first, observes, sequential, backward induction, commitment	§3
proposes, counter-offer, shrinking pie, discount factor δ	§4 (bargaining), §5 (repeated)
types, signals, privately known, prior, "believes"	§6
auction, bid, reserve price, winner's curse, multiple units	§7; securities / debt bids §8
lottery, contest, effort, rent-seeking	§9
match, propose and reject, priorities, school choice, cadre	§10
coalition, core, fair division	§11
utility of wealth, premium, certainty equivalent, gamble	§12
investigate, inspect, search cost, reservation value	§13
elasticity, markup, two plants, natural monopoly	§15
Cournot, Bertrand, Stackelberg, kinked, dominant firm	§16
segments, coupons, student price	§17
entry fee plus per-unit price, membership, subscription	§18
bundle, package, two goods together	§19
advertising-to-sales, joint products, one input two outputs	§20
in-house or buy, transfer price, upstream/downstream	§21
pollution, permits, externality, property rights	§22
non-rival, non-excludable, free riding, basic research	§23
lemons, warranty, monitoring, incentive pay, hidden action	§24

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1 Strategic-form games: dominance, Nash, mixing

1.1 The objects

A **strategic-form game** is three things: a set of players, a set of strategies for each player, and a payoff to each player for every combination of strategies. A **strategy** is a complete plan — in a sequential game it must say what you would do at every node, even nodes you never expect to reach.

Strictly dominant. A strategy s_i is strictly dominant if it gives player i a strictly higher payoff than any of her other strategies *against every possible combination of the others' strategies*. It need not be a good outcome: “Defect” is strictly dominant in the Prisoner’s Dilemma. A strictly dominant strategy is played in *every* Nash equilibrium, because anything else can be improved on.

Weakly dominant. Same comparison but with “at least as good against all, and strictly better against at least one.” Bidding your value in a second-price auction is the standard example. **Careful:** a weakly dominant strategy need *not* be played in every Nash equilibrium, because the opponent’s actual play may make the tie irrelevant — which is exactly why second-price auctions have equilibria in which a low-value bidder wins.

Iterated deletion. Delete strictly dominated strategies, then look again at the smaller game and repeat. The order does not matter for *strictly* dominated strategies. It *does* matter for weakly dominated ones, because deleting one strategy can destroy the tie that made another weakly dominated, so different orders leave different answers. When you use iterated deletion, say which strategy you deleted and why at each round.

Domination by a mixture. A strategy can be undominated by every pure strategy and still be dominated by a *mixture* of them. With payoffs $10, 0 / 0, 10 / 4, 4$ for T, M, B against two columns, B loses to neither T nor M alone, but the half-half mixture of T and M yields 5 against either column, so B is strictly dominated and can be deleted. If a question says “not dominated by any pure strategy, therefore rationalisable,” that is false.

Maximin. Look at the worst payoff each of your strategies could give, and pick the strategy whose worst case is best. This is a *pessimism* criterion; it ignores what the opponent actually wants. A maximin strategy is therefore usually *not* a best response to the opponent’s maximin strategy — the two coincide only in zero-sum games, where the opponent’s interest really is to hurt you. A dominant strategy is always also maximin.

Nash equilibrium. A profile in which each player’s strategy is a best response to the others’. Method: underline each player’s best payoff in each column (row); a cell with all payoffs underlined is an equilibrium. Nash guarantees existence only once mixed strategies are allowed; matching pennies has no pure equilibrium.

Two statements he has set repeatedly. (i) “In a game with a unique Nash equilibrium, the equilibrium maximises the sum of payoffs” — **false**, the Prisoner’s Dilemma is the counterexample. (ii) “If everyone has a strictly dominant strategy, the resulting profile maximises the sum of payoffs” — also **false**, same counterexample.

1.2 Mixed strategies and the indifference principle

If a player mixes between two strategies in equilibrium, she must be *indifferent* between them; otherwise she would put all the weight on the better one. This gives the working rule: **to find player 1’s equilibrium mixture, write down the condition that makes player 2 indifferent.** Your own payoffs pin down the opponent’s mixture, not your own.

The own-payoff puzzle. In asymmetric matching pennies where Row earns 9 at (T, L) and 1 at (B, R) , and Column earns 1 whenever they mismatch: Column’s indifference requires $1 - p = p$, so Row plays T with probability $\frac{1}{2}$ regardless of the 9. Row’s indifference requires $9q = 1 - q$, so $q = 1/10$. Raising Row’s own payoff changes Column’s behaviour, not Row’s. Experiments contradict this, which is what quantal response equilibrium (§14) is designed to fix.

A worked mixture. In Battle of the Sexes with payoffs $(2, 1)$ at (F, F) , $(1, 2)$ at (B, B) and $(0, 0)$ off-diagonal: let the wife play F with probability p . The husband is indifferent when $1 \cdot p = 2(1 - p)$, i.e. $p = 2/3$; symmetrically the husband plays F with probability $1/3$. Each earns $2/3$, which is worse than either pure equilibrium — mixing wastes surplus through miscoordination, here with probability $\frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} = \frac{5}{9}$.

Zero-sum and minimax. In a two-player zero-sum game the mixed equilibrium is exactly the pair of maximin (minimax) strategies, and the equilibrium payoff is the value of the game. Penalty kicks are the standard illustration: the kicker randomises to keep the goalkeeper indifferent, and any predictable pattern is exploited.

1.3 Reading the wording

His statement questions turn on single words. “Must” asks for a claim true in *every* equilibrium. “Can” asks whether *some* equilibrium has the property — a much weaker bar, and usually the answer is yes. “Always” invites a counterexample. “Weakly” versus “strictly” changes the answer in second-price auctions and in iterated deletion. Read the option list twice before answering.

2 Games with discrete strategy sets

He likes games where the strategy set is a finite grid — integer prices, whole-number bids, numbers between 11 and 20. The grid changes the answer, because the usual continuous undercutting argument stops one step short.

2.1 Integer-price Bertrand

Setup. Two (or three) firms sell an identical good and simultaneously post prices. Consumers buy from the cheapest; ties split the demand equally. Market demand is $Q = 10 - P$, each firm's marginal cost is c , and prices must be integers (sometimes restricted to a range).

With *continuous* prices the unique equilibrium is $p = c$ and zero profit, because any common price above cost invites an ε undercut. With *integer* prices the smallest undercut is a whole rupee, which may cost more than it gains, so a price one step above cost survives.

Duopoly, $MC = 2$, $Q = 10 - P$, integer prices. (3, 3) is an equilibrium: each earns $(3 - 2)(7)/2 = 3.5$, and undercutting to 2 earns 0. So the maximum equilibrium profit is 3.5 each. Collusion would set the integer monopoly price: maximise $(P - 2)(10 - P)$ over integers, giving $P = 6$ and joint profit 16.

Duopoly, $MC = 4$, $Q = 10 - P$, integer prices. (5, 5) gives $(5 - 4)(5)/2 = 2.5$ each. No price is even weakly dominant: against a rival at 7, pricing at 6 earns 8 while pricing at 5 earns only 5. If firm A must move first, it sets 5 and B matches, so A guarantees itself 2.5.

Three firms, $MC = 2$, $Q = 10 - P$, integer prices 1–6. There are 14 pure equilibria; the largest total profit in any of them is 7; collusion would earn 16 at $P = 6$. With continuous prices, the lowest price in any equilibrium is exactly $2 = MC$ however many firms there are.

2.2 Sealed-bid auctions with whole-number bids and known values

Setup. Five firms with commonly known values 2, 4, 6, 8, 10 bid natural numbers. In the first-price auction the winner pays her own bid; in the second-price auction she pays the highest rival bid. Ties are broken in favour of the higher valuation.

First price. For a bid b to win in equilibrium, the second-highest-value firm (value 8) must not want to outbid it: $8 - (b + 1) \leq 0$, so $b \geq 7$. Hence the winner is always the top-value firm E, and her best equilibrium payoff is $10 - 7 = 3$ at a bid of 7. No other firm can win, because winning would require bidding at least 10.

Second price. Now the price is somebody else's bid, so a low-value firm can win: if C (value 6) bids 10 and everyone else bids 1, C wins and pays 1. For C to win in equilibrium, E must not want to outbid her and pay C's bid, which needs $10 - b \leq 0$, so C's bid must be at least 10. The best payoff anyone can achieve is E bidding high while all others bid the minimum of 1, earning $10 - 1 = 9$.

Careful: in the first-price auction the top-value bidder *must* win in every equilibrium; in the second-price auction that is false. This pair has been examined at least three times.

2.3 All-pay auction on a grid

Setup. Two bidders compete for a \$100 note. Bids must be multiples of \$5. The high bid wins, ties are broken by a coin, and *both* bidders pay their own bid regardless of who wins.

The best reply to a rival bidding 50 is to bid 55, netting $100 - 55 = 45$. There is **no pure-strategy equilibrium**: from any profile the loser wants to drop to 0 and the winner wants to be undercut-proof, and the cycle never closes. The maximin strategy is to bid 0, which guarantees a payoff of 0 (or 50 if the rival also bids 0). Expected total spending in the mixed equilibrium is close to the prize, which is the rent-dissipation result of §9.

2.4 Traveler's Dilemma

Setup (Basu). Two travellers each write an integer claim between 2 and 100. Both are paid the *lower* claim; in addition the lower claimant receives a bonus of 2 and the higher claimant pays a penalty of 2. If the claims are equal both simply receive that amount.

Maximin is 2: against a rival who claims 2, any claim above 2 pays $2 - 2 = 0$, whereas claiming 2 pays 2. **There is no dominant strategy:** the best reply to 100 is 99, the best reply to 50 is 49, and so on. **The unique Nash equilibrium is (2, 2),** reached by iterated deletion: whatever the rival claims, claiming one less is strictly better, so 100 goes, then 99, and so on down. Real players claim 95–100, which is what level- k reasoning (§14) predicts.

2.5 The 11–20 money-request game

Setup (Arad–Rubinstein). Each player requests an integer number of dollars from 11 to 20 and receives it. A player who requests *exactly one dollar less* than the other receives an extra 20.

Maximin is 20 (it guarantees 20). **There is no pure equilibrium:** any profile invites the one-below deviation. The symmetric mixed equilibrium puts probabilities 25%, 25%, 20%, 15%, 10%, 5% on 15, ..., 20; every request in that support earns exactly 20 in expectation, and 14 earns only 19, which is why the support stops there. Observed play concentrates on 17–19.

The modified version. Add the rule “if both choose the same number they each receive an extra 10.” Now (11, 11) is the unique pure equilibrium, worth $11 + 10 = 21$: the only way to beat it would be to undercut to 10, which is outside the range, and every other request $x \leq 20$ pays just $x < 21$. Every other matched pair (k, k) with $k \geq 12$ fails, because undercutting to $k - 1$ pays $k + 19 > k + 10$.

2.6 Other one-shot classics worth recognising

Beauty contest. Everyone picks a number in $[0, 100]$; the winner is closest to $\frac{2}{3}$ of the average. Iterated deletion drives every rationalisable choice to 0, since no one should ever choose above $\frac{2}{3}$ of the current upper bound. Level-1 picks 33, level-2 picks 22, level-3 picks 15.

Hotelling / median voter. Two sellers choose a position on a line of uniformly spread consumers who buy from the nearer one. Both converge on the **median**. With integer positions 1..100 the equilibrium is (50, 51) or (51, 50). The same argument is the median-voter theorem in politics, and it is why two-party platforms look alike.

War of attrition. Two players want an object worth V_i ; time runs continuously and each loses one unit of payoff per unit of time until somebody concedes. In *every* Nash equilibrium **one player concedes immediately**: if both planned to wait, the player who would eventually give in does strictly better by giving in at once and saving the waiting cost. The valuations determine *who* concedes across the different equilibria, but never *whether* there is delay.

Weakest link, threshold and volunteer games. In a weakest-link game the outcome depends on the *minimum* effort, so every common effort level is an equilibrium and the efficient one is risky to aim for. In a volunteer’s dilemma someone must pay a small cost for everyone to benefit; the symmetric mixed equilibrium has each person volunteering with a probability that *falls* as the group grows, so the chance that nobody volunteers rises with group size — the bystander effect.

3 Sequential games and backward induction

3.1 Method

In a finite game of perfect information, start at the *last* decision node, work out what that player does, replace the node by the resulting payoffs, and repeat. What survives is the **subgame-perfect equilibrium (SPE)**: a Nash equilibrium whose strategies are optimal in every subgame, including those never reached. Its whole purpose is to delete *incredible threats*.

Careful: a strategy in an extensive game must specify an action at every one of that player’s nodes, even ones ruled out by her own earlier choices. Questions that ask “how many strategies does player 2 have?” are counting these complete plans, not single actions.

Entry and the incredible threat. An entrant chooses Enter or Stay out; the incumbent then Fights or Accommodates. Payoffs: (Enter, Accommodate) = (10, 10); (Enter, Fight) = (−5, 2); (Stay out, ·) = (0, 12). There are two Nash equilibria, but “Fight” loses to “Accommodate” once entry has happened ($2 < 10$), so the only SPE is **(Enter, Accommodate)**. The 2022 version with profits 100 / 50 each / −100 each has the same answer: **enter and accommodate**.

Making the threat credible. If the incumbent can first *sink* an investment in capacity that lowers its marginal cost of fighting, fighting becomes a genuine best response, and the entrant stays out. Commitment works by removing your own options, which is why it must be irreversible and observable.

3.2 King of the Hill (2024 final, Q2)

Setup. There is an incumbent King who makes no decisions. Every other player is a Subject and may “Charge the Hill.” Final payoffs: King at the end = 8, Subject at the end = 4, Dethroned King = 0. Once you have been King you make no further decisions.

Line version: subjects move one at a time in a fixed order, each observing what happened before.

Ring version: in each round every remaining subject simultaneously decides to charge or stay idle; if at least one charges, one of them is chosen at random to become the new King and the old King is dethroned; the game ends when a round passes with nobody charging, or after N rounds.

Answer for both versions: the incumbent is dethroned if and only if N is odd.

Line. Work backwards. The last subject always charges, because becoming King at the end pays 8 against 4 for staying a subject and there is nobody left to dethrone him. Subject k therefore charges exactly when the number of subjects *after* him is even, since then the throne will be taken and retaken an even number of times and he keeps it. The parity of N decides whether the first charge ever happens.

Ring. In round N everyone charges, so whoever is King going into that round is dethroned. In round $N - 1$, therefore, everyone stays idle — becoming King now guarantees being dethroned next round, which pays 0 against 4 for idling. In round $N - 2$ everyone charges again, because the new King survives the idle round $N - 1$. The alternation gives the same parity answer.

3.3 Race and take-away games

Setup. A pile of N objects; players alternate removing between 1 and k objects; whoever takes the last object wins.

The player to move **loses** exactly when N is a multiple of $k + 1$, because from such a pile every move hands the opponent a non-multiple, and from a non-multiple you can always return to a multiple. With $k = 2$ the losing piles are 0, 3, 6, 9, ...; from $N = 7$ the winning move is to take 1 and leave 6. Say the invariant explicitly in your answer — that is what earns the marks, not the arithmetic.

3.4 Other backward-induction setups

Pirates dividing loot. Five pirates in rank order propose a split of 100 coins; a proposal passes on a majority (the proposer votes), otherwise the proposer is thrown overboard and the next proposes. Solve from two pirates upwards: with two, the senior takes everything. With three, the most senior buys the cheapest vote with one coin. The pattern is that the proposer needs $\lceil n/2 \rceil$ votes and buys each for one coin more than that pirate would get next round, so the first proposer keeps almost everything. The general lesson: *proposal power plus a known continuation is worth a great deal.*

Centipede. The pot grows as players alternately pass, but taking now always beats what you get if the opponent takes next. Backward induction says take at the very first node; real players pass for several rounds, which is one of the motivations for the limited-foresight model in §14.

Chain store. An incumbent faces entrants in many towns, one at a time. With a known finite number of towns, backward induction unravels the reputation and the incumbent accommodates everywhere. Add a small probability that the incumbent is a “crazy” type who always fights and reputation revives — a standard question is to explain *why* the finite-horizon logic fails once types are uncertain.

Agenda setting. With majority preferences that cycle (A beats B , B beats C , C beats A), whoever sets the order of pairwise votes picks the winner: to make A win, vote B against C first, then the winner against A . Agenda control is real power; see §1 for the cycle itself.

4 Bargaining

4.1 Finite alternating offers with a shrinking pie

Setup (2020 final). A proposes a split of 1000. If B rejects, B proposes a split of 800. If A rejects that, A proposes a split of 500. If B rejects, B proposes a split of 200. If A rejects the last proposal, A receives 20 and B receives 40. Indifference means acceptance.

Work backwards, one round at a time, always asking *what the responder gets by rejecting*:

Round	Proposer	Pie	Reasoning
5	—	—	Rejection gives A 20, so A accepts anything ≥ 20 .
4	B	200	B offers A exactly 20 and keeps 180.
3	A	500	B would get 180 by rejecting, so A offers 180 and keeps 320.
2	B	800	A would get 320 by rejecting, so B offers 320 and keeps 480.
1	A	1000	B would get 480 by rejecting, so A offers 480 and keeps 520.

So A's first proposal is accepted immediately: **A gets 520, B gets 480**. Delay never actually occurs in equilibrium; the threat of delay is what shapes the terms.

The twist worth remembering. If the final rejection instead gave A 20 and B *zero*, B's payoff is still **480**. The reason is that B is the *proposer* in the last round, so his own fallback never constrains him — only the *responder's* fallback matters at each step. A disagreement payoff binds only for the player who has to be persuaded.

4.2 Discounting and Rubinstein

Let the pie be 1 and let each round of delay multiply payoffs by $\delta \in (0, 1)$.

Rubinstein (infinite horizon). There is a unique SPE, agreement is immediate, and the first proposer receives

$$\frac{1 - \delta_2}{1 - \delta_1 \delta_2}, \quad \text{which is } \frac{1}{1 + \delta} \text{ when } \delta_1 = \delta_2 = \delta.$$

Patience is power: raising your own δ raises your share, raising the opponent's lowers it. As $\delta \rightarrow 1$ the split tends to 50–50 and the first-mover advantage vanishes, because waiting one round costs nothing. With $\delta_1 = 0.9$, $\delta_2 = 0.8$ the first proposer gets $0.2/0.28 \approx 0.714$.

4.3 The Nash bargaining solution

When the protocol is not specified, assume the parties split the surplus above their disagreement payoffs.

The Nash solution maximises the **Nash product** $(u_1 - d_1)(u_2 - d_2)$. On a linear frontier $u_1 + u_2 = S$ this gives

$$u_i = d_i + \frac{1}{2}(S - d_1 - d_2),$$

and with bargaining powers $\alpha, 1 - \alpha$ it gives $u_1 = d_1 + \alpha(S - d_1 - d_2)$.

Derivation, if asked: substitute $u_2 = S - u_1$ and differentiate $(u_1 - d_1)(S - u_1 - d_2)$; the first-order condition is $(S - u_1 - d_2) = (u_1 - d_1)$, which rearranges to the formula. The axioms behind it are Pareto efficiency, symmetry, invariance to rescaling utilities, and independence of irrelevant alternatives. The link to §4.2 (the “Nash programme”) is that Rubinstein’s non-cooperative answer converges to it as $\delta \rightarrow 1$.

Comparative statics. $\partial u_1 / \partial d_1 = \frac{1}{2}$: an improvement of Δ in your own outside option is worth only $\Delta/2$ to you, because half leaks to the other side. $\partial u_1 / \partial d_2 = -\frac{1}{2}$. This is why investing in a credible fallback pays even if you never use it.

Hold-up. If one party must sink a relationship-specific investment *before* bargaining, she anticipates that half the return will be bargained away, so she under-invests. Remedies: long-term contracts, vertical integration, or giving the investing party the bargaining power.

5 Repeated games and collusion

5.1 The grim-trigger condition

Setup. A stage game is played every period with common discount factor δ . Write R for the per-period payoff from cooperating, T for the one-off payoff from defecting while the other cooperates, and P for the punishment payoff of the stage Nash equilibrium, with $T > R > P$.

Under grim trigger (cooperate until someone defects, then play the stage equilibrium forever) cooperation is sustainable when

$$\frac{R}{1 - \delta} \geq T + \frac{\delta P}{1 - \delta} \quad \Longleftrightarrow \quad \delta \geq \frac{T - R}{T - P}$$

Read it: the temptation $T - R$ is a one-period gain, while the loss $R - P$ is forever, discounted. Bigger temptation or harsher punishment moves the threshold in the obvious direction.

5.2 What makes collusion easier or harder

Easier	Harder
Few firms; observable prices; frequent interaction	Many firms; secret discounts; lumpy, infrequent orders
Patient firms (high δ), stable demand	Volatile demand, high interest rates
Similar costs and homogeneous products	Very different costs; complex product ranges
Inelastic total demand (the prize is bigger)	Elastic demand (a price rise loses volume)
A cost advantage over non-members	Easy entry by outsiders
Credible, proportionate punishment	Punishments so severe they are not credible

Careful on two options he has used. “Elastic total demand helps a cartel” is **false**. “Threats of extremely costly military-style retaliation help” is also **false**, because a punishment that ruins the punisher fails subgame perfection.

5.3 Standard numbers

Bertrand cartel with n identical firms. Collude at the monopoly price and share equally: $R = \pi^m / n$. A defector takes the whole market for one period: $T = \pi^m$. Punishment is competitive pricing: $P = 0$. The condition becomes

$$\delta \geq \frac{\pi^m - \pi^m / n}{\pi^m - 0} = \boxed{1 - \frac{1}{n}}$$

so two firms need $\delta \geq \frac{1}{2}$ and ten firms need $\delta \geq 0.9$: **collusion gets harder as the number of firms grows.**

Cournot duopoly with linear demand. Collusion needs $\delta \geq 9/17 \approx 0.53$, because the Cournot punishment payoff is positive rather than zero, which weakens the deterrent relative to Bertrand.

Detection lags. If a defection is only detected after k periods, the defector enjoys T for k periods, so the condition becomes $\frac{R}{1-\delta} \geq T \frac{1-\delta^k}{1-\delta} + \frac{\delta^k P}{1-\delta}$, which requires a higher δ . Slow detection is what kills real cartels.

5.4 Two results to be able to state

Finite horizon unravels. In a Prisoner's Dilemma repeated a known 100 times, the last period has no future, so both defect; knowing that, the same is true in period 99, and so on. Cooperation requires an infinite horizon, an uncertain end date, or incomplete information about types.

Folk theorem. For δ close to 1, *any* payoff profile that gives each player at least her minmax value can be supported as a subgame-perfect equilibrium. So "for δ near 1 only permanent cooperation can be sustained" is **false** — the problem with patient players is an embarrassment of equilibria, not a unique one. Conversely, for δ near 0 only repetition of the stage equilibrium survives.

5.5 Price-match guarantees (2024 final, Q6)

A price-match guarantee promises to match any lower rival price. It reads like aggressive competition and functions as the opposite: it makes punishment *automatic and credible*. A firm that cuts its price no longer steals any customers, because the cut is matched instantly; it merely lowers the price for everybody, today and in the future. The one-shot temptation T collapses towards R , so the grim-trigger condition holds for far lower δ and the high price is sustained. Model answer in about 50 words: *"The guarantee removes the gain from undercutting — rivals match immediately, so the defector captures no extra demand and only reduces the industry price. Since defection no longer pays even in the current period, the collusive price is self-enforcing at much lower discount factors."*

6 Bayesian games, signalling and screening

6.1 Bayes–Nash equilibrium: the four-step recipe

A **Bayesian game** adds *types*: each player privately knows her own type, and everyone shares a prior over the others'. A **strategy** is now a plan for each of your types, written (action if type y , action if type n). A **Bayes–Nash equilibrium (BNE)** is a profile in which *every type* of every player is best-responding.

1. Write each player's strategy as a list, one action per type. With two types and two actions each player has four strategies.
2. Fix the opponent's type-contingent strategy.
3. For each of your types separately, compute the expected payoff of each action, weighting the opponent's actions by the *posterior* probability of the opponent's type.
4. Check that every type is best responding. Only then is it an equilibrium.

Careful: do not average over your *own* type. Your own type is known to you; you optimise type by type.

6.2 Bayesian Cournot

Setup. Two firms choose quantities; each has a privately known cost type (high or low) drawn with known probabilities. Inverse demand is linear.

Each type of each firm has its own first-order condition, in which the rival's quantity is replaced by the *expected* rival quantity. With two types on each side you get four linear equations. The 2023 paper's system was

$$94 - 2q_{1H} = \frac{q_{2H} + q_{2L}}{2}, \quad 97 - 2q_{1L} = \frac{q_{2H} + q_{2L}}{2}, \quad 96 - 2q_{2H} = \frac{3q_{1H} + q_{1L}}{4}, \quad 98 - 2q_{2L} = \frac{3q_{1H} + q_{1L}}{4}.$$

Differencing the first pair gives $q_{1L} = q_{1H} + 1.5$ and the second pair gives $q_{2L} = q_{2H} + 1$; substituting reduces it to two equations with solution $q_{2H} = 32.583$, $q_{2L} = 33.583$, $q_{1H} = 30.458$, $q_{1L} = 31.958$. **The shortcut to remember: subtract the two equations of a player's own types first** — the unknown rival aggregate cancels and you get the gap between your own types immediately.

6.3 Signalling (Spence)

Setup. Workers know their own productivity; firms do not. A worker can acquire education y at a cost that is *lower for the more productive type*. Firms pay a wage that depends on observed education.

Education works as a signal only because of **single crossing**: the high type's cost per unit of signal is lower, so there is a level of education the high type is willing to buy and the low type is not. With wages $w_H > w_L$ and costs $c_H < c_L$

per unit, a threshold y^* separates the types when

$$\frac{w_H - w_L}{c_L} \leq y^* \leq \frac{w_H - w_L}{c_H}$$

— the left inequality stops the low type imitating, the right one keeps the high type participating.

Worked (2017 final). Contracts pay 2000 above a threshold x and 1000 below; education costs $200y$ for the low type and $100y$ for the high type. The low type must not imitate: $1000 \geq 2000 - 200x$, so $x \geq 5$. The high type must still find it worth it: $2000 - 100x \geq 1000$, so $x \leq 10$. Minimum separating threshold is 5. If the low type's cost rose to $400y$, the binding condition becomes $1000 \geq 2000 - 400x$ and the threshold falls to **2.5**: the cheaper it is to keep imitators out, the less education is wasted.

Worked (2018 final, three types). Salaries 8000 / 3500 / 1000 for high / medium / low; education costs (0,0,0) for High School, (1000,2000,4000) for B.Tech, (2000,4000,8000) for MBA, (5000,7000,12000) for PhD (in the order high, medium, low). If High School is the low-quality requirement, the medium requirement must be **B.Tech**: it costs the medium type 2000 to gain 2500, and costs the low type 4000 to gain only 2500. Given that, the high requirement must be the **PhD**.

Signalling is wasteful. In a separating equilibrium the education does nothing for productivity; it exists only to sort. Total surplus would be higher if types were observable, which is why cheap verification (audited reports, certified test scores) beats expensive signals.

6.4 Screening

The mirror image: the *uninformed* party moves first with a **menu**, and each type picks the option designed for it. Insurance deductibles, airline fare classes, software tiers and quantity discounts are all screening. Two constraints run the design: **incentive compatibility** (each type prefers its own option) and **participation** (each type prefers its option to nothing). The standard result is “no distortion at the top”: the high type gets the efficient quantity and earns an *information rent*, while the low type's quantity is distorted downwards to make imitation unattractive.

6.5 Adverse selection and market unravelling

Setup (lemons). Sellers know their car's quality, buyers do not. Buyers pay at most the expected quality of the cars actually on offer, which depends on the price, which determines who offers.

The slide example. Peaches are worth 4 lakh, lemons 2 lakh, and 80% are peaches. A buyer who believed the population mix would pay $0.8(4) + 0.2(2) = 3.6$ lakh. At 3.6 no peach owner sells, so only lemons appear, so buyers pay 2. At 2 only lemons are offered. **No peach ever trades.**

The takeover / used-car formula. The seller's value is $v \sim U[a, b]$ and the good is worth kv to the buyer. A bid B is accepted when $v \leq B$, so the conditional mean is $E[v \mid v \leq B] = \frac{a+B}{2}$ and the buyer's expected profit is

$$\Pr(\text{accept}) \left(k \cdot \frac{a+B}{2} - B \right), \quad \text{optimal } B^* = \frac{a}{2-k} \quad (k < 2).$$

With $a = 0$ the answer is $B^* = 0$: **the market shuts down completely**. With $a > 0$ trade survives because even the worst car is worth something. Two exam instances: $v \sim U[0, 20]$, $k = 1.5$ gives $B^* = 0$; $v \sim U[10, 30]$, $k = 1.5$ gives $B^* = 20$.

Certification that verifies nothing does not help. In the 2024 paper sellers could pay \$5 for an “H certificate” that carried no information. A seller certifies and sells at P only if $v \leq P - 5$, so the average certified car is worth $(P - 5)/2$ to its owner and $0.75(P - 5)$ to a buyer, which is still below P . Both markets unravel to a price of **0** and nobody certifies. Only *verified* certification changes beliefs.

The trade question. A buyer and seller must both agree to trade a car at a pre-set price; a lemon is worth 0 to both, a peach 3000 to the buyer and 2000 to the seller, 50–50. Below 2000 only lemons trade, so the buyer gets nothing for a positive price. At or above 2000 both types trade, but the expected value 1500 is below the price. **The maximum acceptable price is 0** — full unravelling again.

6.6 Remedies

Mandatory pooling (compulsory motor insurance, an individual mandate); *verified* certification and audited disclosure; warranties, which are credible because they cost a good producer less; reputation and ratings, which turn the one-shot problem into a repeated game; and screening menus. Which remedy fits depends on whether the hidden thing is a *type* (adverse selection) or an *action* (moral hazard, §24).

7 Auctions

7.1 The four formats and what they are worth

Format	Rule	Strategic equivalent
English (ascending)	price rises until one bidder remains	second-price
Dutch (descending)	price falls until someone accepts	first-price
First-price sealed bid	highest bid wins and pays own bid	Dutch
Second-price (Vickrey)	highest bid wins, pays second-highest bid	English

Second-price: bidding your value is weakly dominant. Your bid decides only *whether* you win, never *what* you pay. Bidding above your value can only win auctions you would rather lose; bidding below can only lose auctions you would rather win. This holds regardless of risk attitude, the number of bidders, or the value distribution — a point he has tested directly.

7.2 Independent private values with $U[0, 1]$ and n bidders

Auction	Equilibrium bid	Seller's expected revenue
Second price	$b(v) = v$	$\frac{n-1}{n+1}$
First price	$b(v) = \frac{n-1}{n}v$	$\frac{n-1}{n+1}$
All-pay	$b(v) = \frac{n-1}{n}v^n$	$\frac{n-1}{n+1}$

Why the first-price bid is $\frac{n-1}{n}v$: bidding as if your value were v' wins with probability $(v')^{n-1}$ and pays $b(v')$, so you maximise $(v')^{n-1}(v - b(v'))$; imposing $v' = v$ in equilibrium and solving the resulting differential equation gives $b(v) = \frac{n-1}{n}v$, which is also $\mathbb{E}[\text{highest rival value} \mid \text{you are highest}]$. Shading falls as n rises, because the risk of losing grows.

Revenue equivalence. With risk-neutral bidders, independent private values drawn from the same distribution, and an auction that always gives the object to the highest bidder and gives a zero-value bidder zero expected payoff, *every* standard format yields the same expected revenue. So “first price raises more than second price” is **false** under those conditions.

When revenue equivalence fails — and these are the interesting questions: risk-averse bidders (they shade less in first price, so *first price raises more*); asymmetric bidders; affiliated or common values (then *ascending* auctions raise more, because bidders learn from each other — the linkage principle); budget constraints; and **securities rather than cash** (§8).

7.3 Reserve prices and the winner's curse

Optimal reserve. With $v \sim U[0, 1]$ and a seller value of 0, the revenue-maximising reserve is $r^* = \frac{1}{2}$, independent of n . It is set by trading off the chance of no sale against a higher price when the sale happens, and it makes the auction inefficient on purpose: sometimes nobody wins although a buyer values the object.

Winner's curse. In a *common-value* auction, winning is bad news: it means your signal was the most optimistic of all. The correct bid conditions on winning, $\mathbb{E}[v \mid s_i, s_i \text{ is the highest of } n]$, and the shading must **increase with n** , because “I had the highest signal out of 20” is much worse news than “out of 2.” Naive bidders who bid their unbiased signal lose money on average.

7.4 Multi-unit auctions (2022 final, Q9–Q13)

Setup. Two identical units are for sale and $n > 2$ bidders each submit a bid for each unit. Bidder i values the first unit she wins at v_i and the second at $w_i < v_i$. The two highest bids across all bidders win.

Discriminatory (pay-as-bid): each winning unit is paid for at its own winning bid.

Uniform price: every unit is paid for at the highest *rejected* bid.

Vickrey: a bidder winning k units pays the sum of the k highest rejected bids submitted by *others*.

Discriminatory: truthful bidding is neither dominant nor an equilibrium — you always shade, exactly as in a single-unit first-price auction. Answer to “which statement is true”: **none of them**.

Vickrey: bidding (v_i, w_i) is weakly dominant — each bid affects only whether you win that unit, never what you pay, so the second-price logic applies unit by unit.

Uniform price: truthful bidding is **not** weakly dominant. Your *second* bid can become the highest rejected bid and hence set the price you pay on your *first* unit, so you shade it. This is **demand reduction**.

Extreme demand reduction can be an equilibrium: everyone bidding $(v_i, 0)$ means the price collapses to zero and nobody gains by deviating, because raising your second bid only raises your own price.

With unit demand a uniform-price auction is a second-price auction, so bidding v_i is weakly dominant again.

A concrete demand-reduction example. Two units, two bidders with values (10, 8) and (9, 7). Truthful bidding makes the highest rejected bid 8, so each pays 8 and bidder 1 earns 2. If bidder 1 instead bids (10, 0), the highest rejected bid falls to 7; she wins one unit at 7 and earns 3. Shading paid.

8 Auctions with securities (2025 final, Q1–Q5)

This is the newest and hardest block on any of his papers, and it is worth learning properly because it is pure method.

Setup. A project is auctioned to two risk-neutral bidders. Bidder i has a private signal v , and signals are independent $U[0, 1]$. A bidder with signal v believes the project's realised revenue Z is uniform on $[0, 2v]$ — so her expected revenue is exactly v .

Instead of cash, bidders submit a **debt bid** $d \geq 0$: a pledge that the seller keeps all revenue up to d , and the buyer keeps anything above d . Formally the payment is $S(d, z) = \min\{z, d\}$.

The paper hands you the key expression: a type- v bidder who wins and pays according to a debt bid d earns

$$v - \mathbb{E}[\min\{z, d\}] = v - d + \frac{d^2}{4v}.$$

The highest d wins; the winner pays according to the *second-highest* bid. Higher types bid higher.

8.1 Q1–Q2: the weakly dominant bid is $d(v) = 2v$

Use the cash-auction principle: *in a second-price auction of anything, bid the amount that makes your payoff on winning exactly zero.* That is what makes both over- and under-bidding harmless. So set the payoff to zero:

$$v - d + \frac{d^2}{4v} = 0 \Rightarrow 4v^2 - 4vd + d^2 = 0 \Rightarrow (2v - d)^2 = 0 \Rightarrow \boxed{d(v) = 2v}.$$

Proof of weak dominance (write it in the same case structure as the cash proof). Let $d^* = 2v$, let d' be any other bid, and let $\hat{d}$ be the highest rival bid. Note first that the payoff from winning at a price $\hat{d}$ equals $v - \hat{d} + \hat{d}^2/4v = (2v - \hat{d})^2/4v$, which is positive when $\hat{d} < 2v$ and negative when $\hat{d} > 2v$.

- If $d' > d^*$: when $\hat{d}$ is below both, you win either way and pay $\hat{d}$ — no difference. When $\hat{d}$ is above both, you lose either way — no difference. When $d' > \hat{d} > d^*$, bidding d^* loses and earns 0, while bidding d' wins at a price $\hat{d} > 2v$ and earns $(2v - \hat{d})^2/4v$ evaluated *beyond* the zero point, which is a strict loss. So d^* is weakly better.
- If $d' < d^*$: the first two cases are again ties. When $d^* > \hat{d} > d'$, bidding d' loses and earns 0, while bidding d^* wins at a price $\hat{d} < 2v$ and earns a strictly positive amount. So d^* is weakly better.

Since a weakly dominant strategy profile is in particular a Nash equilibrium, **both bidders pledging $d(v) = 2v$ is the Nash equilibrium.**

8.2 Q3: the auction is efficient

Efficiency here means the bidder with the highest signal wins. Since $d(v) = 2v$ is strictly increasing in v , the highest bid is always submitted by the highest type. **Yes, efficient.**

8.3 Q4: debt raises more revenue than cash

In a *cash* second-price auction a risk-neutral bidder with signal v values the project at $\mathbb{E}[Z | v] = v$, so she bids $b(v) = v$ and the winner pays the second-highest signal w . Given that the winner's signal is v , w is uniform on $[0, v]$, so the expected cash payment is $\mathbb{E}[w] = v/2$.

In the *debt* auction the winner pays $\mathbb{E}[S(z, 2w)] = 2w - \frac{(2w)^2}{4v} = 2w - \frac{w^2}{v}$. Taking the expectation over $w \sim U[0, v]$, and using $\mathbb{E}[w] = v/2$ and $\mathbb{E}[w^2] = v^2/3$:

$$\mathbb{E}[\text{payment}] = 2 \cdot \frac{v}{2} - \frac{v^2/3}{v} = v - \frac{v}{3} = \boxed{\frac{2v}{3}} > \frac{v}{2}.$$

The debt auction raises strictly more revenue. The intuition to write down: a debt security is worth more to the seller than its face value suggests, because it pays out in exactly the states where the project succeeds; equivalently, a security whose value rises with the true revenue extracts part of the bidder's informational rent that cash cannot reach.

8.4 Q5: first price has no dominant strategy

Exactly as in cash auctions. In a first-price auction what you pay *is* your own bid, so the trade-off between winning more often and paying more depends on what the rivals do; there is no bid that is best against every rival bid. The equilibrium has to be found by maximising expected payoff. Assuming a symmetric increasing equilibrium $d(\cdot)$, a

type v pretending to be v' wins with probability v' (with two bidders and uniform signals) and solves

$$\max_{v'} v' \left(v - d(v') + \frac{d(v')^2}{4v} \right).$$

That much is the required answer. If you want the full solution, the first-order condition at $v' = v$ simplifies to $d'(v) = 1 - \frac{d(v)}{2v}$, whose solution with $d(0) = 0$ is $d(v) = \frac{2}{3}v$; the winner then pays $\frac{2}{3}v - \frac{(2v/3)^2}{4v} = \frac{5}{9}v$. Unconditionally the second-price debt auction raises $4/9$, the first-price debt auction $10/27$, and either cash auction $1/3$ — so **revenue equivalence fails once bids are securities**, and for debt the second-price format is the better one for the seller.

9 Contests and lotteries

Setup (Tullock contest). n players compete for a prize worth V . Player i spends x_i , which is *not refunded*, and wins with probability $x_i / \sum_j x_j$. This is the model of a lottery where tickets are bought, of lobbying, of litigation, and of R&D races.

Player i maximises $\frac{x_i}{x_i + X_{-i}} V - x_i$. The first-order condition is $\frac{X_{-i}}{(x_i + X_{-i})^2} V = 1$; imposing symmetry ($x_i = x$, $X_{-i} = (n-1)x$) gives

$$x^* = \frac{(n-1)V}{n^2}, \quad \text{total spending} = \frac{(n-1)V}{n}, \quad \text{each player's profit} = \frac{V}{n^2}.$$

Rent dissipation: total spending is a fraction $(n-1)/n$ of the prize and tends to V as n grows — with enough contestants the entire value of the prize is burned. With $n = 3$ and $V = 90$: each spends **20**, the organiser collects **60**, and each contestant expects $30 - 20 = 10$.

Asymmetric two-player version. With values V_1, V_2 the equilibrium efforts are $x_i = \frac{V_i^2 V_j}{(V_i + V_j)^2}$ and the stronger player both spends more and wins more often, but the *underdog* dissipates a larger fraction of his own prize. A player whose value is far below the other's may optimally drop out entirely — always check for that corner.

Group contests. If m identical members of a group share a prize V equally and effort is additive within the group, each member internalises only V/m , so total group effort is $\frac{V}{4m}$ in the symmetric two-group case: **larger groups fight less hard**, the classic free-riding-in-collective-action result.

All-pay auction as the limiting contest. If the highest spender wins for certain rather than probabilistically, you get the all-pay auction of §2.3: in the complete-information case with equal values, expected total spending equals the prize exactly.

Lottery versus auction for allocating a licence. A lottery in which winning probability is proportional to spending is a Tullock contest, so with three bidders valuing a licence at 90 the state collects 60. A first-price auction with complete information and equal values drives the price to 90, so the auction raises more and is also efficient when values differ. A state may still prefer the lottery on distributional grounds (it gives small bidders a chance), for political reasons, or because an auction with very asymmetric bidders is a foregone conclusion and deters participation.

10 Matching and market design

10.1 The model and stability

Setup. Two sides — applicants and firms, students and schools, candidates and cadres. Each agent has a strict preference ranking over the other side (and over being unmatched). A **matching** pairs them up. There is no money.

A matching is **stable** if (i) it is *individually rational* — nobody is matched to a partner worse than being unmatched — and (ii) there is no **blocking pair**: no applicant and firm who are not matched to each other but would both prefer to be. Stability is the right criterion because a blocking pair can simply walk away and re-contract, so an unstable matching does not survive.

10.2 Deferred acceptance (Gale–Shapley)

The algorithm. In each round, every unmatched applicant proposes to her most preferred firm among those that have not yet rejected her. Each firm looks at all proposals *in hand* plus any it is currently holding, tentatively **holds** the best acceptable one, and rejects the rest. Repeat until no rejections occur; then all holds become final.

Theorem 1 (it terminates). Each round at least one applicant is rejected by a firm she never proposes to again, and there are finitely many applicant–firm pairs, so the process ends.

Theorem 2 (the outcome is stable). Suppose applicant a prefers firm f to her assignment. Then a proposed to f earlier and was rejected, which happened only because f was holding someone it preferred. Firms' holdings only improve over time, so f ends up with someone it prefers to a . Hence (a, f) is not a blocking pair, and no blocking pair exists.

Theorem 3 (the proposing side does best). Applicant-proposing DA gives *every* applicant the best partner she has in *any* stable matching, and simultaneously gives every firm its worst stable partner. So the two sides' interests are exactly opposed across the set of stable matchings, and **who proposes matters**.

Theorem 4 (strategy-proofness for proposers). Truth-telling is a dominant strategy for the proposing side (Dubins–Freedman, Roth). Intuition: DA already delivers each proposer her best stable partner, so there is nothing left to gain by lying; a misreport can only cause her to be rejected earlier by someone she could have had.

Theorem 5 (the receiving side can manipulate). A firm can gain by **truncating** its list — declaring an acceptable applicant unacceptable so that the applicant it really wants is not taken by a rival. So DA is not strategy-proof for the receiving side.

Theorem 6 (Roth 1982 impossibility). No stable matching mechanism is strategy-proof for *everyone*. A two-by-two example with opposed preferences is enough to construct a profitable manipulation. This is the single most quotable result in the topic.

Many-to-one. With firms having quotas, hospital-proposing DA remains stable but a hospital with capacity 2 can gain by reporting a smaller quota or a truncated list. The **rural hospitals theorem** says the *set* of agents who go unmatched is the same in every stable matching, which is why rural hospitals stay empty whichever stable mechanism is used.

Roommates. With one side only (everyone ranks everyone), a stable matching *may not exist*: with a preferring b , b preferring c , c preferring a , and all preferring each other to the fourth person, every pairing is blocked. Two-sidedness is what makes Gale–Shapley work.

10.3 Top Trading Cycles and school choice

TTC. Each agent points at the owner of her favourite object (or the school she most wants); each object points at its owner (or highest-priority student). With finitely many nodes there must be a cycle; assign everyone in the cycle what they point at, remove them, and repeat. TTC is **Pareto efficient and strategy-proof**, and in a housing market it produces the *unique core allocation*. Its cost is that it trades priorities between students, so it can create **justified envy**: a student may be beaten to a school by someone with lower priority there.

Boston (immediate acceptance). Round 1 assigns first choices permanently by priority; round 2 assigns remaining seats to those whose second choices remain, and so on. Because a first-round rejection wastes your first-choice priority, it is **not strategy-proof** and it punishes ambitious honest ranking: the safe strategy is to rank a school you will certainly get.

	Stable / no justified envy	Pareto efficient for students	Strategy-proof for students
Student-proposing DA	yes	no	yes
TTC	no	yes	yes
Boston	no	(efficient in a weak sense)	no

The trade-off to state: you cannot have stability and student-side Pareto efficiency at once. Boston's outcomes look efficient only because the strategic ranking has already done the work, and it penalises unsophisticated families.

10.4 The IAS cadre-allocation game (2024 quiz)

When candidates are asked to rank cadres and allocation is by rank-plus-priority, the induced game is a strategic-form game whose strategies are the possible rankings. If preferences and priorities are *common knowledge*, backward reasoning gives a unique pure Nash equilibrium in which candidates rank strategically rather than truthfully. If the priorities are *uncertain*, the same game becomes Bayesian and expected utilities must be computed ranking by ranking. The general lesson — and the marks — come from saying *why* the mechanism is not strategy-proof and which theorem (Roth 1982, or Gibbard–Satterthwaite) rules out fixing it while keeping stability.

11 Cooperative games: the core and the Shapley value

11.1 Why a second solution concept

Nash equilibrium asks only whether a *single* player can gain by deviating. Cooperative game theory asks whether a *group* can gain by walking out together. A **coalitional game** gives each coalition $S \subseteq N$ a worth $v(S)$: the total payoff its members can guarantee themselves on their own and divide as they like.

The core is the set of allocations x with

$$\sum_{i \in N} x_i = v(N) \quad \text{and} \quad \sum_{i \in S} x_i \geq v(S) \quad \text{for every coalition } S.$$

Why the inequality is the right translation of “no coalition has an action all its members strictly prefer”: if $\sum_{i \in S} x_i < v(S)$, the coalition can give each member $x_i + \varepsilon$ with $\varepsilon = (v(S) - \sum_{i \in S} x_i) / |S| > 0$ and all are strictly better off; conversely if all members do strictly better under some feasible action for S , then $\sum_{i \in S} x_i$ must have been below $v(S)$.

Recipe. Write $v(S)$ for every coalition; most are zero. Use $\sum_i x_i = v(N)$ to convert each constraint into a *ceiling* on the players left out: $\sum_{j \notin S} x_j \leq v(N) - v(S)$. The binding constraints are nearly always “everyone except i ,” which gives $x_i \leq v(N) - v(N \setminus \{i\})$.

11.2 The landowner and the workers (2022 final, Q1–Q8)

Setup. A landowner’s estate worked by k workers produces $f(k+1)$, where f is increasing with $f(0) = 0$; there are n workers, and only coalitions containing the landowner can produce anything. Marginal product $f(k) - f(k-1)$ is decreasing.

An allocation is in the core **if and only if** $0 \leq x_i \leq f(n+1) - f(n)$ for every worker, with the landowner taking the residual.

Necessity. Workers alone are worth nothing, so $x_i \geq 0$. The coalition of the landowner with all workers *except* i produces $f(n)$, so $f(n+1) - x_i \geq f(n)$, i.e. $x_i \leq f(n+1) - f(n)$.

Sufficiency. Take the landowner with k workers, worth $f(k+1)$. The $n-k$ excluded workers hold at most $(n-k)[f(n+1) - f(n)]$ between them, and $f(n+1) - f(k+1)$ is a sum of $n-k$ marginal products each at least the smallest one, so the coalition’s share is at least $f(k+1)$. No coalition blocks.

The answers. Maximum to the landowner: $f(n+1)$. Minimum total to the workers: **0**. The most dangerous deviating coalitions: **the landowner with every worker but one**. Maximum to a single worker: $f(n+1) - f(n)$. “Every allocation with $0 \leq x_j \leq f(n) - f(n-1)$ is in the core” is **false**, because decreasing marginal product makes that ceiling too generous. If the workers unionise and ban any group of fewer than n from working with the landowner, the core is *not* every allocation: the landowner can still work alone and produce $f(1)$, so $x_1 \geq f(1)$ survives.

The economics in one line. The core is the set of competitive-looking outcomes: each worker is paid at most the marginal product of the last worker, and the landowner collects the Ricardian rent. Unionisation works by *destroying the landowner’s outside option* — it removes the coalition that was capping wages.

11.3 Three standard core examples

One seller, two buyers (house worth 0 to the seller, 10 and 8 to the buyers). The core is $x_S \in [8, 10]$, $x_{B_1} = 10 - x_S$, $x_{B_2} = 0$: the house sells to the high-value buyer at a price between 8 and 10, and the *losing* bidder is the reason the seller is guaranteed 8 — the same logic as a second-price auction.

Glove market (one left glove, two right; a pair is worth 1). The core gives **everything to the scarce side**: $x_L = 1$. A one-unit change in relative scarcity moves the entire surplus.

Three-person majority game ($v(S) = 1$ whenever $|S| \geq 2$). Adding the three pairwise constraints gives $2\sum x_i \geq 3$ while efficiency says $\sum x_i = 1$. The core is **empty** — the signature of pure majority conflict, and the reason coalition governments are unstable.

11.4 The Shapley value

Line the players up in a random order; each player is paid her **marginal contribution** to the set already present; the Shapley value is the average over all $n!$ orders. It is the unique rule satisfying efficiency, symmetry, the dummy property and additivity.

For three players, write the six orders in a column and fill in three marginal contributions per row — ninety seconds and essentially error-proof.

Game	Shapley value	Core	In the core?
Glove market (L, R_1, R_2)	$\frac{2}{3}, \frac{1}{6}, \frac{1}{6}$	$x_L = 1$	no
Seller with buyers valuing 10 and 8	$\frac{19}{3}, \frac{7}{3}, \frac{4}{3}$	$x_S \in [8, 10]$	no
Landowner, $f = 10, 16, 19$	15, 2, 2	worker cap 3	yes
Three-person majority	$\frac{1}{3}$ each	empty	n/a
Airport runways costing 6, 9, 15	2, 3.5, 9.5	non-empty	yes

The airport shortcut is worth knowing: split each *increment* of cost equally among the players who need it (6 shared by three, then 3 shared by two, then 6 alone). That serial rule *is* the Shapley value of the cost game.

Core versus Shapley in one line each. The core answers “what is stable?” — it may be empty or large, and everything in it is immune to group deviations. The Shapley value answers “what is fair?” — it always exists and is unique, but it

need not be stable. For *convex* games (a player's marginal contribution grows with the coalition she joins) the Shapley value is always in the core.

Power indices. Apply the Shapley formula to a voting game and a player's value becomes the probability she is *pivotal*. With shareholders holding 50, 40 and 10 votes and a quota of 51, the power indices are $\frac{2}{3}$, $\frac{1}{6}$, $\frac{1}{6}$: the 40% holder has exactly as much power as the 10% holder. Power is not proportional to votes.

12 Risk, insurance and the value of information

12.1 Expected value versus expected utility

A risky prospect has two summaries: the expected *value* $\mathbb{E}(X) = \sum_i \text{Pr}_i X_i$ and the expected *utility* $\mathbb{E}u(X) = \sum_i \text{Pr}_i u(X_i)$. They rank prospects differently whenever u is not linear, and essentially every question in this topic turns on the difference.

Careful: $\mathbb{E}[u(X)]$ is the average of the utilities; $u(\mathbb{E}X)$ is the utility of the average. Jensen's inequality says $\mathbb{E}[u(X)] \leq u(\mathbb{E}X)$ for concave u , and that inequality is the definition of risk aversion. Writing $\sqrt{0.5(4) + 0.5(16)}$ instead of $0.5\sqrt{4} + 0.5\sqrt{16}$ throws away the whole question.

Attitude	Behaviour	Utility
Risk averse	prefers the sure mean to the gamble	concave: $\sqrt{x}$, $\ln x$, $x^{1/3}$
Risk neutral	indifferent; only the mean matters	linear: x
Risk loving	prefers the gamble	convex: x^2 , e^x

12.2 Certainty equivalent and risk premium

The **certainty equivalent** CE solves $u(\text{CE}) = \mathbb{E}[u(X)]$; the **risk premium** is $\mathbb{E}[X] - \text{CE}$, the expected value the person will give up to avoid the risk.

The thirty-second rule. For $u = \sqrt{\cdot}$: take square roots of the outcomes, average them, square the result — $\text{CE} = (\mathbb{E}\sqrt{X})^2$. For $u = \ln x$, CE is the geometric mean $e^{\mathbb{E}\ln X}$. For $u = x^{1/3}$, $\text{CE} = (\mathbb{E}X^{1/3})^3$. For $u = x^2$ (risk loving), $\text{CE} = \sqrt{\mathbb{E}X^2}$.

More risk aversion means a lower CE and a higher risk premium. Among the CRRA family $x^{1-\rho}/(1-\rho)$, a smaller exponent is more risk averse, so $x^{1/3}$ is *more* risk averse than $x^{1/2}$ — which is exactly what the 2025 insurance question exploits.

12.3 Insurance: the three premiums

Setup. Wealth W ; a loss L occurs with probability π ; utility u is increasing and concave. Full insurance pays L in the loss state in return for a premium P paid in *both* states, leaving certain wealth $W - P$.

- Actuarially fair premium** $= \pi L$, the insurer's expected payout. This is the competitive price, because many independent customers make the insurer effectively risk neutral (the variance of the average claim is $\sigma^2/n \rightarrow 0$).
- Maximum premium the customer accepts** $= W - \text{CE}$, where $u(\text{CE}) = (1 - \pi)u(W) + \pi u(W - L)$. She buys iff $u(W - P) \geq \mathbb{E}u$, i.e. iff $P \leq W - \text{CE}$.
- Monopoly insurer** who knows u : charges the maximum (or the largest permitted amount just below it).

The gap between (2) and (1) is the **risk premium**, and it is the gain from trade that makes insurance exist.

Two partial-coverage results. At an actuarially fair price a risk-averse person buys *full* coverage. At a loaded price ($p > \pi$) she buys strictly *less* than full coverage — which is why real policies carry deductibles.

Careful: a more risk-averse customer supports a *higher* premium only when the insurer has market power. Under perfect competition the premium is πL and does not depend on risk aversion at all.

12.4 Worked numbers to recognise

Situation ($u = \sqrt{\cdot}$ unless stated)	$\mathbb{E}u$	CE	Max prem.	Fair prem.
House 100, 10% fire loss of 50	9.707	94.23	5.77	5
House 36L, 50% fall to 16L	5	25L	11L	10L
Store 64L, 20% theft to 16L	7.2	51.84L	12.16L	9.6L
Necklace 1,000,000, 20% total theft	800	640,000	360,000	200,000
Wealth 400, 25% loss of 300	17.5	306.25	93.75	75
Investment: 1,000,000 w.p. 0.6, else 250,000	800	640,000	—	$\mathbb{E}X = 700,000$

The last row is the classic trap: the most a risk-averse buyer should pay for the asset is its certainty equivalent 640,000, *not* its expected value 700,000.

A risk-loving twist. With $U(W) = W^2$, wealth 1m, and a fair coin paying +500,000 or replacing the necklace with 500,000, participation with fee F (in millions) requires $0.5(1.5 - F)^2 + 0.5(0.5 - F)^2 \geq 1$, i.e. $F^2 - 2F + 0.25 \geq 0$, so $F \leq 1 - \frac{\sqrt{3}}{2} \approx 0.134$ m. A risk lover will pay for a fair gamble, but not much.

12.5 Two customers, one contract, and paying for data (2025 final, Q12)

Setup. Jiwan and Rajpal each have wealth 900 and each faces a 50% chance of losing 800. Jiwan's utility is $W^{1/2}$ and Rajpal's is $W^{1/3}$, but you do not know which is which. You are risk neutral and may make take-it-or-leave-it full-coverage offers; you may not offer two different deals to the same person.

Step 1 — each person's maximum premium. Jiwan: $\mathbb{E}u = 0.5\sqrt{100} + 0.5\sqrt{900} = 0.5(10) + 0.5(30) = 20$, so $CE = 400$ and $P_J^{\max} = 900 - 400 = 500$. Rajpal: $\mathbb{E}u = 0.5(100)^{1/3} + 0.5(900)^{1/3} = 0.5(4.6416) + 0.5(9.6549) = 7.1482$, so $CE = 7.1482^3 = 365.26$ and $P_R^{\max} = 900 - 365.26 = 534.74$. Rajpal's cube-root utility is more concave, so he will pay more — state that as the sanity check.

Step 2 — one common deal (part a). Your expected payout per customer is $0.5(800) = 400$.

- Charge 500: *both* accept, profit = $2(500 - 400) = 200$.
- Charge 534.74: only Rajpal accepts, profit = $534.74 - 400 = 134.74$.

So offer a premium of 500 and earn 200. This is the pooling-versus-skimming trade-off of §24: a single price must be acceptable to the *least* willing customer you want to keep.

Step 3 — buying the data (part b). With tailored deals you charge each his own maximum:

$$\text{profit} = 500 + 534.74 - 2(400) - 20 = 214.74 > 200.$$

So **yes, pay the 20**; the information is worth $234.74 - 200 = 34.74$ gross and 14.74 net. The general principle is in §12.6: information has value only when it *changes what you do*.

12.6 The value of information

$$\text{EVPI} = \underbrace{\mathbb{E}[\max_a \text{payoff}(a, \text{state})]}_{\text{decide after learning}} - \underbrace{\max_a \mathbb{E}[\text{payoff}(a, \text{state})]}_{\text{decide before}} \geq 0.$$

Worked. A project pays +100 with probability 0.6 and −80 otherwise; the alternative pays 0. Without information: $\mathbb{E} = 60 - 32 = 28 > 0$, so invest. With perfect information: invest only in the good state, $\mathbb{E} = 0.6(100) = 60$. So $\text{EVPI} = 32$, which equals the avoided loss 0.4×80 . **Imperfect information is worth strictly less:** with an 80%-accurate signal the same project's information is worth only 15.

13 Search: investigating before choosing

Setup (Weitzman's Pandora problem). Each closed box i costs c_i to open and contains a random prize X_i . You may open boxes in any order, stop at any time, and take the best prize you have found.

Give each box a **reservation value** r_i defined by $c_i = \mathbb{E}[\max(X_i - r_i, 0)]$. Then: **Selection rule** — open the unopened box with the highest r_i . **Stopping rule** — stop as soon as the best prize in hand is at least the highest remaining r_i .

For a “prize V with probability p , else nothing” box, $c = p(V - r)$, so $r = V - \frac{c}{p}$.

Read the formula: the prize shrunk by the cost *inflated* by the odds of ever collecting it. A box with a huge prize and a tiny probability can still have a high reservation value.

Worked (2024 final, Q1). Project 1 is a known winner worth \$1; project 2 pays \$21 with probability 0.09; project 3 pays \$10 with probability 0.76; investigation costs \$1 each.

$$r_2 = 21 - \frac{1}{0.09} = 9.89, \quad r_3 = 10 - \frac{1}{0.76} = 8.68.$$

Both exceed the \$1 in hand and $r_2 > r_3$, so **open project 2 first**. If it wins, stop and take it ($21 > r_3$). If not, open project 3; take it if it wins, else take project 1. Expected payoff

$$0.09(21 - 1) + 0.91(0.76)(10 - 2) + 0.91(0.24)(1 - 2) = 1.8 + 5.5328 - 0.2184 = 7.1144,$$

against 7.032 if project 3 is opened first and 6.84 if only project 3 is opened. **The point of the question:** project 3 has four times the expected value (7.6 versus 1.89) yet goes second, because you open first the box whose success would most decisively *end* the search. Say that sentence if asked why.

Reservation wage. Offers are $U[0, 100]$ and each draw costs c . Accept the first offer above w^* where $c = \mathbb{E}[\max(w - w^*, 0)] = \frac{(100 - w^*)^2}{200}$; with $c = 2$, $w^* = 80$. A higher search cost lowers w^* ; a more dispersed offer distribution raises it; unemployment benefits act like a lower c and lengthen spells.

Secretary problem. Candidates arrive in random order and must be accepted or rejected on the spot. Reject the first $n/e \approx 37\%$, then accept the first candidate better than all seen. This picks the best with probability about $1/e \approx 37\%$, and that probability does not fall as n grows. The shape — a learning phase then a threshold phase — is the answer whenever a question says “no recall, no second chances.”

14 Other solution concepts he may define in the paper

Correlated equilibrium. A mediator draws a profile of recommendations from a publicly known distribution and tells each player only *her own*. It is an equilibrium if each player, given what her recommendation implies about the others', prefers to obey. In Chicken with payoffs $(0, 0)$, $(7, 2)$, $(2, 7)$, $(6, 6)$, recommending (C, C) , (D, C) , (C, D) with probability $\frac{1}{3}$ each yields 5 per player, beating the mixed Nash payoff $14/3$. The gain comes from the signals being *private*; if the draw were announced publicly only Nash equilibria would survive.

Evolutionarily stable strategy. s is an ESS if a small group of mutants playing any $t \neq s$ does strictly worse. In Hawk–Dove with prize V and fighting cost $C > V$ there is no pure ESS and the mixed ESS has a hawk share $p^* = V/C$ (so $V = 4$, $C = 6$ gives two-thirds hawks). **Careful:** if $C < V$ the formula exceeds 1 and the answer is the pure ESS “always Hawk.”

Trembling-hand perfection. Require each strategy to stay optimal when opponents make small mistakes. In two-player games this removes equilibria that use weakly dominated strategies — it is the normal-form way of saying “do not believe incredible threats.”

Quantal response equilibrium. Players choose better actions more often but not always: $\Pr(a) = e^{\lambda U(a)} / \sum_b e^{\lambda U(b)}$, with U computed against the others' choice probabilities, which are generated the same way. $\lambda = 0$ is uniform randomness, $\lambda \rightarrow \infty$ is Nash. Unlike Nash, QRE produces an **own-payoff effect**: in the asymmetric matching pennies of §1, $\lambda = 1$ gives $\Pr(T) \approx 0.89$ rather than 0.5. For two actions, $\Pr(a) = 1 / (1 + e^{-\lambda[U(a) - U(b)]})$.

Level- k . Level-0 randomises or picks the salient option; level- k best-responds to level $k - 1$. Guessing $\frac{2}{3}$ of the average: 33, 22, 15 (Nash 0). Guessing half the average: 25, 12.5, 6.25. The 11–20 game with level-0 at 20: 19, 18, 17. **Always state your level-0 assumption** — it is the only place a reasonable answer can differ, and it usually carries a mark.

Limited foresight (his own 2022 paper). A player with foresight k sees only the next k stages and values nodes beyond her horizon by a rule of thumb (the average of the best and worst payoffs still reachable). Do backward induction on the *truncated* tree, replacing horizon nodes by the stated value, and show the truncation explicitly. The model rationalises passing in centipede games, delay in bargaining, and risk-taking against inexperienced opponents.

Cursed equilibrium. A fully cursed player predicts others' *average* behaviour but ignores the link between their actions and their information. In the takeover game with $v \sim U[0, 20]$ and $k = 1.5$, a cursed buyer values the firm at $1.5 \times 10 = 15$ regardless, bids $\arg \max_{\frac{B}{20}} (15 - B) = 7.5$, and actually loses about 0.70 per attempt. The diagnostic question is always: *conditional on my offer being accepted, what do I now know?*

Social preferences. Fehr–Schmidt inequity aversion, $U_i = x_i - \alpha \max(x_j - x_i, 0) - \beta \max(x_i - x_j, 0)$. In the ultimatum game the responder accepts a share s iff $s \geq \alpha / (1 + 2\alpha)$, which is $1/3$ when $\alpha = 1$. A proposer with guilt weight $\beta > \frac{1}{2}$ offers an equal split outright. This explains 40–50% offers, positive dictator giving, and why costly punishment sustains public-good contributions.

Braess's paradox. With 4000 drivers, routes $S \rightarrow X$ ($T/100$ minutes) then $X \rightarrow E$ (45), and $S \rightarrow Y$ (45) then $Y \rightarrow E$ ($T/100$), the equilibrium is 2000 on each route taking 65 minutes. Open a free instantaneous road $X \rightarrow Y$ and everyone takes $S \rightarrow X \rightarrow Y \rightarrow E$, taking $40 + 0 + 40 = 80$ minutes — worse for everybody, and still an equilibrium because deviating costs 85. Adding capacity can slow a congested network.

15 Monopoly and market structure

15.1 Marginal revenue and the markup rule

Revenue is $R = P(Q)Q$, so differentiating as a product,

$$MR = P + Q \frac{dP}{dQ} = P \left(1 + \frac{1}{E} \right) = P \left(1 - \frac{1}{|E|} \right).$$

Selling one more unit brings in P on that unit but forces a price cut on every unit already sold; the more elastic demand is, the smaller that cut and the closer MR is to P .

$MR > 0 \iff |E| > 1$, and revenue is maximised where $|E| = 1$. Since a monopolist sets $MR = MC > 0$, it **always** operates on the elastic part of demand. Setting $MR = MC$ gives the **Lerner index**

$$\frac{P - MC}{P} = \frac{1}{|E|}, \quad P = \frac{MC}{1 - 1/|E|}.$$

Uses. A firm charging 10 with $MC = 8$ faces $|E| = 1/0.2 = 5$. A firm charging 600 with $|E| = 2.5$ has $MC = 600(0.6) = 360$; if it charges 900 to a second group, that group's elasticity is $|E|$ with $1 - 1/|E| = 0.4$, i.e. **1.67**.

Careful: a higher Lerner index means a higher percentage markup, *not* higher profit — profit also depends on the level of demand and on fixed costs. And with iso-elastic demand, more elastic demand forces a *lower* markup; profits cannot be ranked from elasticity alone.

15.2 The linear workhorse

With $P = a - bQ$ and constant $MC = c$: $MR = a - 2bQ$ (same intercept, twice the slope), so

$$Q^* = \frac{a - c}{2b}, \quad P^* = \frac{a + c}{2}, \quad \pi^* = \frac{(a - c)^2}{4b}, \quad DWL = \frac{(a - c)^2}{8b}.$$

The monopolist sells exactly *half* the competitive quantity, and the deadweight loss is half the profit. **Always invert the demand curve before differentiating** — the commonest arithmetic slip in the topic.

15.3 Three comparative statics he tests

1. **A change in fixed cost changes nothing.** Fixed costs do not enter MC , so price and quantity are unchanged and only profit falls. A licence fee shifts average total and average fixed cost, leaving marginal and average *variable* cost alone.
2. **A per-unit tax t raises the price by $t/2$** with linear demand and constant cost, because MC becomes $c + t$ and $P^* = (a + c + t)/2$. The firm absorbs the other half.
3. **A monopolist has no supply curve.** A supply curve maps price to quantity independently of demand; a monopolist's quantity depends on the *shape* of demand through MR , so the same price can go with different quantities.

The correct pricing algorithm is: find Q^* from $MR = MC$, then read the price off the *demand curve* at Q^* . It is *not* “average cost plus a markup,” not “where $|E| = 1$ ” (that maximises revenue), and not “ $P - MC = 1/|E|$ ” (the Lerner index is a ratio, not a difference).

15.4 Multi-plant monopoly

At the optimum $MC_1(q_1) = MC_2(q_2) = MR(Q)$: if one plant's marginal cost were lower, shifting a unit there would cut cost at unchanged revenue. Practically, **horizontally sum the plants' marginal cost curves** (for each level of marginal cost add the quantities) and intersect with MR .

Constant marginal costs are a corner. If $MC_1 = c_1 < c_2 = MC_2$ are both constant, equality is impossible, so **produce everything in the cheaper plant** and set $MR = c_1$; the expensive plant makes **zero**. He builds option lists that omit 0 precisely to punish an automatic “split it” answer.

Worked examples. $Q = 10 - P$, plants at constant 2 and 3: $10 - 2Q = 2$ gives $Q = 4$, $P = 6$, and the costly plant makes **0**. The deadweight-loss-minimising price ceiling is $P = MC = 2$.

$Q = 100 - P$, plants at 10 and 12: $Q = 45$ at the cheap plant, **0** at the other, $P = 55$.

$MC_A = 8Q_A$, $MC_B = 6Q_B$, $MR = 21 - 2Q$: with a common marginal cost m , $Q_A = m/8$ and $Q_B = m/6$, so $Q = 7m/24$ and $m = 24Q/7$; setting $21 - 2Q = 24Q/7$ gives $Q = 147/38$ and $Q_B = m/6 = 42/19 \approx 2.21$.

$Q = 20 - P$, $C_a = 4Q_a + 2Q_a^2$, $C_b = 2Q_b + 4Q_b^2$: the first-order conditions $16 = 6Q_a + 2Q_b$ and $18 = 2Q_a + 10Q_b$ give $Q_a = 2.214$, $Q_b = 1.357$; check that $MC_a = MC_b = MR = 12.857$.

15.5 Natural monopoly and regulation

A **natural monopoly** has average cost declining over the whole relevant range — large fixed costs and low marginal cost, so $MC < AC$ throughout. One firm then serves the market more cheaply than two, so splitting it forfeits the scale economies. Railways, transmission grids, pipelines.

Ordinary monopoly: a price ceiling where MC meets demand removes the whole deadweight loss; a lower ceiling creates a shortage.

Natural monopoly: $P = MC$ is efficient but lies below AC , so it needs a subsidy. **Average-cost pricing** ($P = AC$ where AC crosses demand) keeps the firm solvent with a much smaller deadweight loss.

Worked. $Q = 800 - P$ with cost $3.5Q^2$: $AC = 3.5Q$ is *rising*, so marginal-cost pricing works — $800 - Q = 7Q$ gives $Q = 100$, $P = 700$, and $AC = 350 < 700$, so the firm profits. With cost $1000Q - 2Q^2$: $AC = 1000 - 2Q$ and

$MC = 1000 - 4Q$ both decline, so it is a natural monopoly. Marginal-cost pricing gives $Q = 66.7$, $P = 733$, but $AC = 867 > P$: not viable. Average-cost pricing solves $800 - Q = 1000 - 2Q$, giving $Q = 200$ and $P = 600$.

15.6 Monopolistic competition

Many firms, differentiated but close-substitute products, free entry. In the short run each sets $MR = MC$ and may profit; in the long run entry shifts each firm's demand left until it is **tangent** to average cost. At the tangency $P = AC$ (zero economic profit) and, because demand slopes down, the tangency is on the *falling* part of AC . Hence two inefficiencies — $P > MC$ (deadweight loss) and **excess capacity** (output below minimum- AC scale) — offset by the benefit of product variety.

Zero profit does not mean zero deadweight loss. Deadweight loss comes from the gap between price and *marginal* cost; zero profit is about the gap between price and *average* cost. Both hold simultaneously at the tangency, and also for a natural monopoly regulated to $P = AC$. He has set this three times, for monopoly and for monopolistic competition, and the answer is **false** both times.

Also long-run monopolistically competitive output is *not* at minimum average cost, and price exceeds both the short-run and long-run minimum average cost.

15.7 Barriers to entry

Structural: scale economies relative to market size, patents and licences, control of an essential input, network effects, heavy sunk advertising or R&D. **Strategic:** excess capacity that makes a price war credible, limit pricing, product proliferation, exclusive contracts, a fighting reputation. Modern antitrust adds consumer data and privacy to the classical consumer-price standard; macroeconomic inflation is not an antitrust objective.

16 Oligopoly

The four models differ in exactly two dimensions: what firms choose, and when.

	Simultaneous	Sequential
Choose quantity	Cournot	Stackelberg (<i>first-mover</i> advantage)
Choose price	Bertrand	Price leadership (<i>second-mover</i> advantage)

16.1 Cournot

With $P = a - bQ$ and constant costs, firm 1 maximises $[a - b(q_1 + q_2) - c_1]q_1$, giving the **reaction function** $q_1 = \frac{a - c_1 - bq_2}{2b}$, which slopes *down*. Solving both:

$$q_i = \frac{a - 2c_i + c_j}{3b}; \quad \text{if } c_1 = c_2 = c: q_i = \frac{a - c}{3b}, P = \frac{a + 2c}{3}, \pi_i = \frac{(a - c)^2}{9b}.$$

With n symmetric firms, $q_i = \frac{a - c}{(n + 1)b}$ and $P = \frac{a + nc}{n + 1} \rightarrow c$: Cournot interpolates between monopoly and perfect competition.

Careful: the asymmetric formula can come out negative, in which case the true equilibrium is a corner — the expensive firm produces zero and the other is an unconstrained monopolist (check that the monopoly price is below the excluded firm's cost).

Worked. $P = 850 - 0.5Q$ with costs 150 and 350: reactions $q_A = 700 - 0.5q_B$ and $q_B = 500 - 0.5q_A$ give (600, 200), total **800** and $P = 450$. Maximum *joint* profit is achieved by producing everything at the low cost: $850 - Q = 150$ gives $Q = 700$, $P = 500$, profit = 245,000 — more than the 200,000 the two earn in Cournot, but it requires a side payment because one firm produces nothing.

Cartel instability in numbers. With $P = 20 - Q$ and $MC = 4$, collusion sets $q = 4$ each for a profit of 32. If the partner sticks to 4, my best response is $BR(4) = 6$, giving $P = 10$ and profit 36 > 32. Cheating pays in the one-shot game; \$5 is what stops it.

16.2 Stackelberg

The leader substitutes the follower's reaction function into its own profit and maximises. With linear demand and equal costs the leader produces the *monopoly* quantity $\frac{a-c}{2b}$ and the follower half of that, so $Q = \frac{3(a-c)}{4b}$, $P = \frac{a+3c}{4}$, and profits are $\frac{(a-c)^2}{8b}$ and $\frac{(a-c)^2}{16b}$.

The advantage comes from commitment, not information. The leader's quantity is deliberately *not* a best response to the follower's quantity — that is the point, and it is a favourite true/false trap. Two statements worth distinguishing:

- "The follower best-responds to the leader's quantity, and the leader incorporates that reaction" — **true** (a correct description of backward induction).

- “The leader’s quantity is a best response to the follower’s quantity” — **false**.

16.3 Bertrand and why the paradox breaks

With a homogeneous product, equal constant costs c , unlimited capacity and simultaneous prices, the unique equilibrium is $p_1 = p_2 = c$ with zero profit. *Proof:* any common price above c invites an ε undercut that doubles your market; any price below c loses money. Two firms suffice for the competitive outcome, and the result is unchanged with three firms — the *lowest* price in any equilibrium is exactly c .

Five escapes, each a possible question: unequal costs (the low-cost firm prices just below the rival’s cost); **discrete prices** (§2.1); capacity constraints (Kreps–Scheinkman: price competition after capacity choice reproduces Cournot); product differentiation (below); and repetition (§5).

16.4 Differentiated Bertrand

With $q_i = \alpha - \beta p_i + \gamma p_j$, profit $(p_i - c)q_i$ gives $p_i = \frac{\alpha + \beta c + \gamma p_j}{2\beta}$, a reaction function that slopes up: prices are **strategic complements**.

Worked (2018). $Q_D = 100 - P_D + P_K$, zero cost. The first-order condition $100 - 2P_D + P_K = 0$ and symmetry give $P = 100$ each, $Q = 100$, profit 10,000 — equal prices but *positive* profits, so the Bertrand paradox is gone. If Danti leads: Kanti’s reaction is $P_K = (100 + P_D)/2$, so Danti maximises $P_D(150 - P_D/2)$, giving $P_D = 150$, $P_K = 125$, quantities 75 and 125, and profits 11,250 and 15,625.

Resolve “first or second mover?” with the three numbers, not the slogan. In the standard linear model *both* firms earn more under sequential pricing than under simultaneous pricing; it is only the *comparison between them* that favours the follower. So “moving first reduces profits” is wrong, and “the leader gets the larger share” is also wrong. In 2018 the marked answer was that posting first is a good idea because it *increases* Danti’s profit.

Worked (2022). $Q_t = 44 - 2P_t + P_g$, $MC = 2$. Simultaneous: $P_t = (48 + P_g)/4$ and symmetry give $P = 16$, profit 392 each. Tablet leading: maximise $(P_t - 2)(56 - 1.75P_t)$, so $P_t = 17$ and $P_g = 16.25$, with Tablet’s profit 393.75. Tablet would pay up to $393.75 - 392 = 1.75$ to move first. Consumer surplus is highest under **simultaneous** pricing, because prices are lowest.

The general statement. Quantities are strategic substitutes (reaction curves slope down), so committing to a *large* quantity makes the rival retreat — commitment is tough and the first mover gains. Prices are strategic complements (curves slope up), so committing to a high price invites the rival to raise price too, which helps the leader but helps the follower more.

16.5 Dominant firm with a competitive fringe

Residual-demand method. (1) Subtract fringe supply from market demand at each price to get the dominant firm’s residual demand. (2) Invert it and write the residual marginal revenue (same intercept, twice the slope if linear). (3) Set residual $MR = MC_{\text{dominant}}$ and read the price off the *residual demand* curve. (4) The fringe supplies the rest; check it is non-negative.

OPEC. $Q = 100 - P$, fringe $Q^C = 3P$, OPEC cost $C(q) = q$: residual $q = 100 - 4P$, so $P = 25 - q/4$ and $MR = 25 - q/2 = 1$, giving $q = 48$ and $P = 13$ (fringe supplies 39; total $87 = 100 - 13$). *Leader with rising cost.* $P = 150 - 3Q$, fringe supply $P = 60 + 3Q$, leader $MC = 2q$: residual $q = 70 - \frac{2P}{3}$, so $P = 105 - 1.5q$ and $MR = 105 - 3q = 2q$, giving $q = 21$ and $P = 73.5$.

16.6 Kinked demand (Sweezy)

Rivals match price *cuts* but not *increases*, so demand is elastic above the current price and inelastic below. The marginal revenue curve is therefore built branch by branch and has a **vertical gap** at the kink quantity. Any marginal cost inside the gap leaves price and quantity unchanged — **price rigidity**.

Worked. Above $P = 8$, $Q = 10 - P$ so $MR = 10 - 2Q$, which is 6 at the kink $Q = 2$; below, $P = 24 - 8Q$ so $MR = 24 - 16Q$, which is -8 at $Q = 2$. The gap is $[-8, 6]$. With $MC = 8 > 6$ the firm is on the upper branch: $10 - 2Q = 8$ gives $Q = 1$ and $P = 9$. Moving MC from 6 to 1 stays inside the gap, so the price change is 0. A second version with the lower branch $Q = 6 - 0.5P$ gives a gap of $[4, 6]$, so marginal costs of 5 and 4 both leave $Q = 2$ and the change is again 0.

16.7 The comparison table

With $P = a - Q$, two firms, common cost c :

	Total output	Price	Profits
Monopoly / perfect collusion	$(a - c)/2$	$(a + c)/2$	$(a - c)^2/4$ total
Stackelberg	$3(a - c)/4$	$(a + 3c)/4$	leader $(a - c)^2/8$, follower $(a - c)^2/16$
Cournot	$2(a - c)/3$	$(a + 2c)/3$	$(a - c)^2/9$ each
Bertrand / competition	$a - c$	c	0

Output rises and price falls down the table; industry profit falls. Check with $a - c = 16$: collusion 32 each, Stackelberg 32 and 16, Cournot $256/9 \approx 28.4$ each, Bertrand 0.

17 Price discrimination

17.1 The three degrees and the three conditions

A single price leaves money in two places: surplus with consumers who would have paid more, and unserved consumers who value the good above marginal cost. Price discrimination attacks one or both.

- **First degree (perfect):** a different price for every unit to every buyer, equal to willingness to pay.
- **Second degree:** prices differ by the *quantity* bought — block tariffs, quantity discounts, menus. The firm cannot tell who is who, so buyers **self-select**.
- **Third degree:** the firm observes a *characteristic* (student, senior, region, coupon user, time) and charges each observable group a different price.

It requires **market power** (impossible under perfect competition), **identifiable differences** in willingness to pay, and **no resale**.

17.2 First degree

Under perfect discrimination the price cut applies only to the marginal buyer, so $MR = P$ and the demand curve *is* the marginal revenue curve. Consequences: output rises to the **competitive level**, **total surplus equals the competitive level** and there is **no deadweight loss**, and the firm captures all of it — consumer surplus is zero. It is efficient and maximally unfair, and that combination is the point.

Worked. $Q = 10 - P$ and $C = Q^2$. Single price: $10 - 2Q = 2Q$ gives $Q = 2.5$, $P = 7.5$, profit **12.5**. Perfect discrimination: $10 - Q = 2Q$ gives $Q = 10/3$ and profit is the whole area between demand and marginal cost, $\int_0^{10/3} (10 - 3q) dq = 16.67$.

Imperfect first degree is the practically important case: doctors and lawyers charging by ability to pay, negotiated car prices, B2B pricing by “value in use” to each client, personalised online pricing, and *auctions* (which reveal willingness to pay but usually leave the winner some surplus). Pricing *client by client* is first degree; pricing broad observable groups is third degree — the phrase “each different client” versus “each segment” is what decides it.

First degree is always the upper bound. “Successful third-degree discrimination typically generates more revenue than successful first-degree” is **false**.

17.3 Third degree

Maximising $R_1(Q_1) + R_2(Q_2) - C(Q_1 + Q_2)$ gives $MR_1 = MR_2 = MC$, and substituting $MR_i = P_i(1 - 1/|E_i|)$ gives the **inverse-elasticity rule**

$$\frac{P_1}{P_2} = \frac{1 - 1/|E_2|}{1 - 1/|E_1|} \implies \text{the less elastic segment pays more.}$$

Method. If MC is *constant* the segments decouple: solve $MR_i = MC$ separately. If MC *rises*, they interact: either write both first-order conditions with $Q = Q_1 + Q_2$ and solve simultaneously, or build the horizontally summed MR curve and intersect it with MC .

Worked (rising cost). $Q_1 = 10 - P_1$, $Q_2 = 24 - 4P_2$, $C = Q^2$. Then $MR_1 = 10 - 2Q_1$, $MR_2 = 6 - Q_2/2$ and $MC = 2Q$. Setting each equal to $2Q$: $Q_1 = 5 - Q$ and $Q_2 = 12 - 4Q$; adding, $Q = 17/6$, so $Q_1 = 13/6$, $Q_2 = 4/6$ and $Q_1 - Q_2 = 1.5$. The aggregate MR curve is $Q = 17 - 2.5m$, so at $Q = 2$ the marginal revenue is $m = 6$.

Worked (constant cost). Foreign $P = 10 - Q$, domestic $P = 20 - 2Q$, $C = 4Q$. Domestic: $20 - 4Q = 4$ gives $Q = 4$ and $P = 12$; foreign: $10 - 2Q = 4$ gives $Q = 3$ and $P = 7$.

Coupons. Coupon users have $|E| = 3$ and others $|E| = 1.5$, so $\frac{P_c}{P_r} = \frac{1 - 1/1.5}{1 - 1/3} = \frac{1/3}{2/3} = \frac{1}{2}$: a **50 per cent** discount. The self-selection logic is that people willing to clip coupons are exactly the price-sensitive ones.

Corner solutions. A segment is served only if its **choke price** exceeds marginal cost at the optimum; otherwise its formula quantity comes out negative and you must re-solve with it excluded. Example: $Q_1 = 30 - P_1$, $Q_2 = 8 - P_2$, $MC = Q$. Assuming both are served gives $Q = 9.5$ and $Q_2 < 0$, so serve only segment 1: $Q_1 = 10$, $P_1 = 20$, $MC = 10$ — and segment 2's best customer values the good at 8, below 10. Note that a *flatter* demand curve (*b* large) says nothing about the choke price a/b , so “if the firm sells in the steeper segment it must sell in the flatter one” is **false**.

Two more verdicts. Optimal two-segment discrimination *weakly beats* a single price, because charging both segments the same price is one of the available schemes (a nesting argument you can reuse everywhere). And third-degree discrimination leaves a deadweight loss in *every* served segment, since $P_i > MC$.

17.4 Intertemporal and peak-load pricing

Intertemporal discrimination sells the same good dear to impatient buyers first and cheaply later — hardcover then paperback, new phone then price cut, opening-weekend tickets. Costs are the same in both periods; only buyers differ. It is *not* about producing a stock first and releasing it over time.

Peak-load pricing applies when demand differs across periods and capacity is scarce, so the marginal cost of serving a peak customer genuinely is higher. Electricity, tolls, hotels in season. Because the cost differs, it is not really discrimination, and setting $P = MC$ separately in each period is what a *regulator* would do — it raises efficiency.

Airlines' free random seat versus paid seat selection is risk aversion plus third-degree discrimination: the free option is a gamble, and those who dislike it sort themselves into paying.

17.5 Ranking the pricing strategies (2025 final, Q8)

From best to worst *for the seller*, assuming all the data were available:

1. **First-degree price discrimination** — charges each unit at its buyer's maximum willingness to pay, so the seller takes the entire consumer surplus. Nothing can beat it, so it is the upper bound.
2. **Two-part tariff** — captures *all* the surplus when consumers are identical (set the per-unit price at marginal cost and the fee at the whole consumer surplus), but only part of it when types differ, because a single entry fee must satisfy the weakest customer served.
3. **Third-degree price discrimination** — one price per observable group, so prices are only loosely tied to individual willingness to pay and surplus is left within each group.
4. **A single per-unit price** — least responsive of all to willingness to pay, so it leaves the most surplus with consumers.

Two caveats worth one sentence each if you have room. The ranking of (2) and (3) can reverse when consumers are very heterogeneous, since a two-part tariff suffers from heterogeneity while third-degree discrimination feeds on it. And (2) beats (4) always, because setting the fee to zero reproduces single pricing.

18 Two-part tariffs

A two-part tariff charges a fixed **entry fee** T plus a **per-unit price** P : total payment $T + Pq$. Amusement parks, gyms, mobile plans, printers and cartridges, franchise contracts. The per-unit price controls how much surplus is *created*; the fee controls how much the firm *keeps*.

18.1 Identical consumers

Set $P^* = MC$ and $T^* = CS(MC)$. *Why*: the firm's total take is $CS(P) + (P - MC)q(P)$, which is exactly **total surplus** at price P , and total surplus is maximised at $P = MC$. Make the pie as big as possible, then eat all of it.

Worked. $Q = 10 - p$, $MC = 5$: set $P = 5$, so $q = 5$ and $T = \frac{1}{2}(10 - 5)(5) = 12.5$, with no deadweight loss. Simple monopoly pricing would give $Q = 2.5$, $P = 7.5$ and profit 6.25 — exactly half.

18.2 Heterogeneous consumers: the algorithm

1. Fix which set of types you will serve (you must compare all candidate sets).
2. For a given P , the fee is limited by the *weakest customer you intend to keep*: $T(P) = \min_{i \text{ served}} CS_i(P)$.
3. Write $\pi(P) = (\# \text{ served}) T(P) + (P - MC)(\text{total quantity of served types})$ and maximise over P . Watch for **kinks**, where the identity of the lowest-surplus type switches — check the kink price as well as the interior first-order condition.
4. Compare across candidate sets.

Worked (two types, $MC = 0$). $Q_1 = 10 - p$ and $Q_2 = 4 - p/4$, so $CS_1 = \frac{1}{2}(10 - p)^2$ and $CS_2 = (16 - p)^2/8$, which are equal at $p = 4$. For $p < 4$ the fee is limited by type 2 and profit is $\frac{(16-p)^2}{4} + p(14 - \frac{5p}{4})$, whose derivative $6 - 2p$ vanishes at $p^* = 3$, giving $T = 21.125$ and profit **73**. For $p \geq 4$ profit falls throughout, and serving one type alone yields at most 50. **With heterogeneous consumers the optimal usage price is strictly above marginal cost**, because raising P takes money from the big user at little cost in fee revenue.

18.3 Exclusion is often optimal

Worked (2018). One million customers each of $Q_1 = 100 - P$ and $Q_2 = 160 - P$, zero marginal cost. Serving both: $\pi = (100 - p)^2 + p(260 - 2p)$, maximised at $p = 30$ with $\pi = 10,900$ million. Serving only the high type: $\pi = \frac{1}{2}(160 - p)^2 + p(160 - p)$ has derivative $-p \leq 0$, so $p^* = 0$, $T = 12,800$ and $\pi = 12,800$ million. **Serve only the high type.**

Worked (YogShala). 100 Yogis with $P = 1000 - 2Q$ and 50 Zens with $P = 500 - Q$, zero cost. Serving both (fee = Zen surplus) gives $\pi = 75(500 - p)^2 + p(75,000 - 100p) = 18,750,000 - 25p^2$, so $p = 0$ and $\pi = 18.75\text{m}$ with a fee of 125,000. Serving only the Yogis gives $\pi = 25,000,000 - 25p^2$, so $p = 0$ and a fee of $\frac{1}{2}(1000)(500) = 250,000$ with $\pi = 25\text{m}$. **Exclude the Zens.**

18.4 The statement questions, settled

Claim	Verdict
An optimal two-part tariff weakly beats optimal single pricing	true (nesting)
Given the optimal P , the fee equals the lowest surplus <i>among the types served</i>	true
For any given P , the revenue-maximising fee is the minimum surplus over <i>all</i> types	false (exclusion)
The optimal tariff leaves all types with positive surplus	false (the marginal served type gets zero)
Serving all types always beats serving a subset	false
Raising the entry fee raises entry-fee revenue	false (fewer customers enter)

The general profit function to keep in mind is $\pi = n(T)T + (P - MC)Q(n(T), P)$ with $n'(T) < 0$.

Menus of plans. Offering several fee-and-allowance combinations lets each type self-select, converting one tariff into a second-degree screening menu (§6.4) and recovering much of what heterogeneity destroyed. That is why real mobile and gym contracts are menus.

19 Bundling and tying

19.1 Why bundling works

Selling two goods separately is limited by the *dispersion* of each reservation price. Selling only the package changes the relevant distribution to that of the *sum*, which is far less dispersed when the two valuations are **negatively correlated**, so one bundle price captures nearly all of it.

Worked. Distributor A values two films at 15 and 5; distributor B at 12 and 6. Separately: film 1 at 12 sells two copies for 24, film 2 at 5 sells two for 10, total **34**. The bundle values are 20 and 18, so a bundle priced at 18 sells to both for **36**. **The control:** change B's second value to 4 so the valuations run in the same direction — separate pricing gives 32 and the bundle gives 32, so bundling gains nothing. Correlation, not bundling itself, creates the gain.

19.2 What a consumer does under mixed bundling

She picks the largest *non-negative* surplus among: nothing (0); good 1 only ($r_1 - p_1$); good 2 only ($r_2 - p_2$); the bundle ($r_1 + r_2 - p_B$); and both separately ($r_1 + r_2 - p_1 - p_2$, relevant only if $p_1 + p_2 < p_B$, which a sensible firm avoids).

Claim	Verdict
If $r_1 + r_2 > p_B$ she buys the bundle	false — a single good can give more surplus
If $r_1 + r_2 < p_B$ she buys nothing	false — she may still buy one good
If $r_1 < p_1$ and $r_2 < p_2$ she buys nothing	false — the bundle can still beat nothing
Anyone who can afford the bundle buys it	false
The bundle is bought only if it is cheaper than buying both separately	true

Worked. Entrée 1500, dessert 500, both for 1800; a diner values them at 1900 and 100. Surpluses: 400 (entrée), -400 (dessert), 200 (bundle). She buys **the entrée only**, although her total value 2000 exceeds the bundle price. *Worked.* $p_1 = 10$, $p_2 = 8$, $p_B = 16$. “Good 1 alone is best” requires $r_1 - 10 \geq r_1 + r_2 - 16$, i.e. $r_2 \leq 6$ — the bundle adds the second good for an extra 6, so anyone valuing it above 6 takes the bundle.

19.3 Discrete optimisation

Worked (four consumers, integer prices). Reservation prices (30, 120), (60, 90), (90, 60), (120, 30); indifferent consumers buy. With cost 40 per unit per good, set $p_1 = p_2 = 120$ and $p_B = 150$: the extreme consumers buy one good each for a margin of 80, the middle two buy the bundle for a margin of 70, total **300**. (That figure relies on ties breaking in the firm's favour; pricing at 119 gives a robust 298.) With zero cost, all four bundle values equal 150, so a pure bundle at 150 sells to everyone: **600**.

Worked (a continuum). If $r_{\text{onion}} + 2r_{\text{tomato}} = 10$, the bundle value is $r_o + r_t = 10 - r_t$ with $r_t \in [0, 5]$, so values run over $[5, 10]$ and the highest price at which *everyone* buys is the minimum, **5**.

19.4 Continuous independent values (2025 final, Q11)

Setup. A monopolist in both markets faces 10,000 consumers. Each consumer's reservation value for an OTT subscription is $r_1 \sim U[0, 100]$ and for auto insurance is $r_2 \sim U[0, 100]$, drawn *independently*. Costs are zero, so revenue is profit.

(a) Selling separately. A consumer buys good 1 iff $r_1 \geq P_1$, which happens with probability $1 - P_1/100$. Revenue per consumer is $P_1(1 - P_1/100)$, maximised at $P_1 = 50$ with revenue 25. The same for good 2, so total per consumer is 50 and overall **500,000**.

(b) Pure bundling. A consumer buys iff $r_1 + r_2 \geq P$. Draw the 100×100 square: the region where $r_1 + r_2 < P$ is a triangle, so for $P \leq 100$,

$$\Pr(r_1 + r_2 \geq P) = 1 - \frac{1}{2} \left(\frac{P}{100} \right)^2.$$

Revenue per consumer is $R(P) = P - \frac{P^3}{2(100)^2}$, and $R'(P) = 1 - \frac{3P^2}{2(100)^2} = 0$ gives

$$P^* = \sqrt{\frac{20,000}{3}} = 81.65, \quad R = 54.43 \text{ per consumer,} \quad \text{total } 544,331.$$

For $P \geq 100$ the buying region is the opposite corner triangle, $\frac{1}{2} \left(\frac{200-P}{100} \right)^2$, and the derivative $\frac{1}{2}(200-P)(200-3P)$ is negative throughout, so the best point on that branch is $P = 100$ with 500,000. **Pure bundling beats separate selling by about 9%.**

Why it works here even though the values are independent: bundling does not need negative correlation to help, only *lower dispersion of the sum*. The sum of two independent uniforms is triangular — concentrated near 100 — so a single price near 82 captures almost everyone.

(c) Mixed bundling does better still. Offer the bundle at P_B and each good alone at p , with $P_B < 2p$. A consumer buys good 1 alone iff $r_1 \geq p$ and $r_2 \leq x$, where $x \equiv P_B - p$ is the *incremental* price of the second good; she buys the bundle iff $r_1 \geq x$, $r_2 \geq x$ and $r_1 + r_2 \geq P_B$. In units of 100, with $a = p/100$ and $b = x/100$, revenue per consumer is

$$100 \left[a \cdot 2b(1-a) + (a+b) \left((1-b)^2 - \frac{(a-b)^2}{2} \right) \right].$$

Yes, it beats pure bundling. The scheme in his own key — $P_B = 81.65$ with singles at $p = 71.65$ (so $x = 10$) — yields about **546,800**, a gain of roughly 2,500 over pure bundling, which is the “ ≈ 0.25 per consumer” his answer quotes. Optimising properly gives $p \approx 66.7$ and $P_B \approx 86.2$, worth about **549,200**.

The economics to write down. Pure bundling forces a consumer who loves one good and is indifferent to the other either to pay for both or buy nothing. Mixed bundling peels those consumers off with a single-good price, collecting revenue from people the bundle would have lost, while the incremental price x keeps everyone in the middle buying the package. The cost is that some consumers who *would* have bought the bundle now buy one good and pay less; mixed bundling wins when the first effect dominates, which it does here.

19.5 Tying

Making the purchase of one good conditional on buying another: printers and cartridges, razors and blades, franchise inputs. Three motives: **metering** (consumable usage measures intensity of demand, so a cheap machine plus expensive consumables is a disguised two-part tariff), **foreclosure** (leveraging power into an adjacent market — the competition-law concern), and **quality protection**. The “Barbenheimer” combined ticket is the cinema version of bundling, and it helps exactly when tastes for the two films are negatively correlated.

20 Advertising and joint products

20.1 The advertising-to-sales ratio

With demand $Q(P, A)$ and advertising spend A , profit is $PQ - C(Q) - A$. The first-order condition in A is $(P - MC) \frac{\partial Q}{\partial A} = 1$: the last rupee of advertising must generate one rupee of extra contribution margin. Multiplying by $A/(PQ)$ and using the markup rule $\frac{P-MC}{P} = \frac{1}{|E_P|}$,

$$\boxed{\frac{A}{PQ} = \frac{E_A}{|E_P|}}$$

where E_A is the advertising elasticity. The ratio **rises** with advertising elasticity and **falls** with price elasticity — more elastic demand means a thinner margin, so each extra unit sold is worth less and advertising is worth less. With $E_A = 0.2$ and $|E_P| = 4$, spend 5% of revenue.

Two statements. “The optimal ratio is increasing in the price elasticity” is **false**; “increasing in the advertising elasticity” is **true**. And if two products have the same advertising elasticity but cement’s demand is *less* price elastic than noodles’, cement’s ratio is *higher*, so a claim that it is lower is **false**.

20.2 Joint products in fixed proportions

Setup. One process yields one unit of each of two outputs: a barrel of oil gives petrol and diesel, a unit of milk gives butter and buttermilk. You choose only how many units of the joint input to process, then a price in each output market.

Build total marginal revenue by **vertically adding** the two marginal revenue curves and produce where it equals MC. But **never sell a unit whose marginal revenue is negative** — dispose of it instead. So

$$MR_{\text{total}}(Q) = \max(MR_1(Q), 0) + \max(MR_2(Q), 0),$$

which makes the curve **kinked** wherever one product’s MR hits zero. Then read each price off *its own* demand curve at the quantity actually sold, which may differ across products.

Worked (disposal binds). $Q_1 = 10 - P_1$ gives $MR_1 = 10 - 2Q$ (zero at $Q = 5$); $Q_2 = 20 - 0.5P_2$ gives $MR_2 = 40 - 4Q$ (zero at $Q = 10$). So $MR_{\text{total}} = 50 - 6Q$ up to $Q = 5$ and $40 - 4Q$ beyond. With $MC = Q$: at the kink $MR = 20 > 5$, so continue; $40 - 4Q = Q$ gives $Q = 8$. Process 8, sell only 5 units of product 1 at $P_1 = 5$ (dispose of 3), and all 8 units of product 2 at $P_2 = 24$. With $MC = 19Q$ instead: $50 - 6Q = 19Q$ gives $Q = 2$, below the kink, so no disposal and $P_1 = 8$, $P_2 = 36$.

Worked (butter and buttermilk). $Q = 100 - P$ and $q = 80 - p$ with processing $MC = 20$: $MR_{\text{total}} = 180 - 4Q = 20$ gives $Q = 40$, exactly where the buttermilk MR reaches zero, so no disposal. Butter price = 60, buttermilk = 40.

Worked (petrol and diesel). $Q_d = 10 - P_d$ and $Q_p = 10 - P_p/4$ with cost $7.5Q^2$: $MR_d = 10 - 2Q$, $MR_p = 40 - 8Q$, so $MR_{\text{total}} = 50 - 10Q = 15Q$ gives $Q = 2$. Then $P_d = 8$ and $P_p = 32$. **Label both prices explicitly** — the printed key for this question reported the petrol price when the diesel price was asked for, so naming them protects you.

Worked (the disposal test). $Q_p = 10 - P_p$ and $Q_d = 16 - 4P_d$, so $MR_p = 10 - 2Q$ and $MR_d = 4 - Q/2$. At $Q = 6$, $MR_p = -2$ and $MR_d = +1$. The negative one is truncated to zero because the firm would dispose of that unit, so total marginal revenue is 1. This single idea is the most testable thing in the topic.

21 Transfer pricing and vertical integration

21.1 Set-up and the master rule

An upstream division makes an input at cost $C_u(Q_1)$; a downstream division turns one unit of input into one unit of final good at extra cost C_d and sells where the firm has market power.

Net marginal revenue is the downstream division’s willingness to pay for one more input unit:

$$NMR(Q) = MR_{\text{final}}(Q) - MC_{\text{downstream}}(Q).$$

At the optimum, $NMR(Q_c) = (\text{opportunity value of a unit}) = MC_u(Q_1)$, and the correct **transfer price is the upstream division’s marginal cost at the optimal total output** — in all three cases below. Pricing the internal transfer above marginal cost causes **double marginalisation**.

21.2 Case 1: no outside market for the input

The opportunity value is zero, so simply $NMR(Q) = MC_u(Q)$ with $Q_1 = Q_c$.

Worked (logs). $P = 20,000 - Q$, upstream $MC = 4Q$, downstream $MC = 8000$: $NMR = 12,000 - 2Q = 4Q$ gives $Q = 2000$, transfer price $4(2000) = 8000$, furniture price 18,000. *Worked (geysers).* $P_g = 10,000 - Q$, coil cost Q^2 so $MC_u = 2Q$, assembly 5000: $NMR = 5000 - 2Q = 2Q$ gives $Q = 1250$, transfer price 2500.

21.3 Case 2: a perfectly competitive outside market

A unit can always be bought or sold at P_e , so the opportunity value is P_e and **the two divisions decouple**:

$$MC_u(Q_1) = P_e \text{ (how much to make),} \quad NMR(Q_c) = P_e \text{ (how many final goods to sell),}$$

and the firm buys $Q_c - Q_1$ if positive, or sells $Q_1 - Q_c$ if negative. **The transfer price is P_e .**

Make or buy is a false choice. With strictly increasing marginal cost starting from zero, the first units are cheaper to make than to buy and the last are dearer, so the firm does **both** — making up to $MC_u = P_e$ and buying the rest. Any option asserting a corner is wrong.

Worked (logs at $P_e = 7800$). $4Q_1 = 7800$ gives $Q_1 = 1950$ made; $12,000 - 2Q_c = 7800$ gives $Q_c = 2100$ used, so the firm *buys* 150. At $P_e = 8400$: $Q_1 = 2100$ and $Q_c = 1800$, so it *sells* 300 on the market. *Worked (coils).* At $P_e = 2000$: $Q_1 = 1000$, $Q_c = 1500$, so **buy 500 coils** and sell 1500 geysers. At $P_e = 4000$: $Q_1 = 2000$, $Q_c = 500$, so **sell 1500 coils**.

The net-position threshold. The firm switches from net buyer to net seller where $Q_1 = Q_c$, and that threshold price is always *equal to the Case-1 transfer price*. For the coils, $\frac{P}{2} = \frac{5000-P}{2}$ gives $P = 2500$, and indeed the no-market answer was $Q = 1250$ with a transfer price of 2500. Use this as a free consistency check.

Writing the profit function (2025-style). With Q_1 made in house, Q_2 bought (negative if sold), $Q = Q_1 + Q_2$ assembled and sold, assembly cost c_A per unit and demand $Q = K - BP$:

$$\pi = \left(\frac{K - Q}{B} \right) Q - c_A Q - P_e Q_2 - C(Q_1).$$

The first term is revenue from the inverted demand curve; the second the assembly cost; the third the net cost of market transactions (a purchase if $Q_2 > 0$, a sale if $Q_2 < 0$); the fourth in-house production cost. The two first-order conditions are exactly $C'(Q_1) = P_e$ and $NMR(Q) = P_e$.

A corner question worth knowing. If in-house cost is Q_e^2 so that $MC_u = 2Q_e$ is zero at zero output, and $P_e > 0$, then the firm always makes a positive number of units in house — “produces 0 engines in-house” is **impossible**. Selling zero engines, buying zero engines, or selling zero final goods are all possible.

21.4 Case 3: market power in the outside input market

The firm is then a multi-market monopolist selling the same input into two markets, so

$$MR_e(Q_e) = NMR(Q_c) = MC_u(Q_e + Q_c),$$

which is the third-degree discrimination logic with the internal division as one “segment” whose demand curve is NMR .

Worked. $P_E = 6000 - Q_E$, $P_C = 8000 - Q_C$, engine cost Q^2 , conversion cost 2000. Profit is $(6000 - Q_E)Q_E + (8000 - Q_C)Q_C - 2000Q_C - Q^2$ with $Q = Q_E + Q_C$. The first-order conditions are $6000 - 2Q_E - 2Q = 0$ and $6000 - 2Q_C - 2Q = 0$, so $Q_E = Q_C$ and $Q_E = 1000$, $Q = 2000$. Check: $MR_E = 4000$, $NMR = 4000$, $MC_u = 4000$. *Worked.* $P_E = 8000 - Q_E$, $P = 20,000 - Q_C$, assembly $10,000Q_C$, engine cost q^2 : the same method gives $Q_E = 1000$, $Q_C = 2000$, $q = 3000$, transfer price 6000 and engine market price 7000.

21.5 Double marginalisation

If upstream and downstream are *separate* monopolists, the upstream firm marks up over its cost and the downstream firm marks up again over that, so the final price is higher and output lower than an integrated firm would choose — worse for consumers *and* for the two firms combined.

Numbers. $P = 100 - Q$, zero costs. Integrated: $MR = 100 - 2Q = 0$ gives $Q = 50$, $P = 50$, profit 2500. Separate: the downstream firm facing wholesale price w sets $Q = (100 - w)/2$, which is the upstream firm’s demand; upstream maximises $w(100 - w)/2$, so $w = 50$, then $Q = 25$ and $P = 75$, with profits $1250 + 625 = 1875 < 2500$. **Fixes:** vertical integration with a transfer price at marginal cost; a two-part wholesale tariff (franchise fee plus per-unit price at cost); or resale price maintenance. All three restore the integrated outcome.

22 Externalities and permits

22.1 Efficiency and the Pigouvian remedy

With $MSC = MPC + MEC$ and $MSB = MPB + MEB$, efficiency requires $MSB = MSC$ while the market delivers $MPB = MPC$.

- **Negative externality:** $MSC > MPC$, so the market produces **too much**.
- **Positive externality:** $MSB > MPB$, so the market produces **too little**.

A **Pigouvian tax equal to MEC evaluated at the efficient quantity Q^*** (or a subsidy equal to $MEB(Q^*)$) aligns private with social incentives. Note it is the marginal external cost at Q^* , not at the market quantity.

The statement he reuses: “If a good has positive externalities, marginal social benefit exceeds marginal private benefit, leading to too much of the good being produced” — **false**, it leads to too little. The companion statement, that assigning property rights helps when bargaining is costless, is **true**.

Worked. Supply $Q = P$, demand $Q = 300 - P$, marginal external cost $= Q$. Then $MSB = 300 - Q - Q = 300 - 2Q$ and $MSC = Q$, so $Q^* = 100$ (the market would give 150). The optimal tax is $MEC(100) = 100$: with it, demand becomes $Q = 200 - P$ and meets supply at 100. **If an external cost is given as a function without the word “marginal”, state in one clause which reading you are using.**

22.2 Instruments

Efficient abatement sets $MCA = MEC$ — zero emissions is almost never efficient.

- **Emission standard:** simple, but a uniform cap forces high-cost abaters to do as much as low-cost ones, which is needlessly expensive.
- **Emission fee:** every firm abates until $MCA = \text{fee}$, so marginal abatement costs are equalised — the condition for achieving any given total at minimum cost.
- **Tradeable permits:** issue permits equal to the target and let firms trade. In equilibrium every firm's MCA equals the permit price, so the cost-minimising allocation is reached **without the regulator knowing any firm's costs**. That informational property is the whole point, and it is the marked answer whenever he asks about the class permit game.

Fees versus standards under cost uncertainty. If MEC is steep relative to MCA (a threshold effect, a toxic pollutant), getting the *quantity* wrong is very costly, so prefer a **standard**. If MEC is flat (a gradually accumulating stock pollutant like CO_2), prefer a **fee**.

Worked (why trading beats a uniform standard). Three firms each emit 3 units and get 1 permit, so 6 units must be abated. Marginal abatement costs for the first, second and third unit are A: 2, 4, 6; B: 5, 8, 12; C: 10, 15, 20. The six cheapest units are 2, 4, 5, 6, 8, 10, so A abates 3, B abates 2, C abates 1, at a total cost of 35. The permit price lies between the last unit abated (10) and the cheapest unit not abated (12), so $P \in [10, 12]$; A sells its permit and C buys one. A uniform standard of 2 each costs $6 + 13 + 25 = 44$ — 26% more for the same abatement.

22.3 The class permit game in general

Each player draws k independent marginal abatement costs uniformly on $[0, 100]$ (one per unit of current emissions) and receives m free permits. A unit is abated iff its cost is below the price, which happens with probability $P/100$, so market clearing requires $k \cdot \frac{P^*}{100} = k - m$:

$$P^* = 100 \times \frac{k - m}{k}$$

Checks: $k = 4, m = 2$ gives 50; $k = 2, m = 1$ gives 50; $k = 4, m = 1$ gives 75.

The 2025-style derivation (two units, one permit). At price P , classify firms: both costs below P (probability $(P/100)^2$) means abate both, emit nothing, **sell** the permit; exactly one below (probability $2 \cdot \frac{P}{100} \cdot \frac{100-P}{100}$) means abate one, emit one, no trade; both above (probability $(\frac{100-P}{100})^2$) means abate nothing, emit two, **buy** one. Setting expected supply equal to expected demand,

$$\left(\frac{P}{100}\right)^2 = \left(\frac{100-P}{100}\right)^2 \Rightarrow P = 100 - P \Rightarrow P^* = 50.$$

Intuition to add: a firm is equally likely to be a buyer or a seller only when the price sits at the middle of the cost distribution.

22.4 Property rights and Coase

Coase theorem. If property rights are clearly assigned and bargaining is costless, the parties reach the **efficient** outcome regardless of *who* holds the right; the assignment determines only the *distribution* of surplus.

Worked (steel and fisheries). Payoffs (steel, fishery): (Filter, Treat) = (300, 300); (Filter, Not treat) = (300, 500); (No filter, Treat) = (500, 200); (No filter, Not treat) = (500, 100). No filter dominates for steel, so the Nash equilibrium is (No filter, Treat) with a total of 700. The efficient profile is (Filter, Not treat) with 800. If steel holds the right to pollute, the fishery pays it to filter and they split the gain of 100, giving (550, 250); if the fishery holds the right to clean water, the outcome is (300, 500) directly. Efficient either way; only the split changes.

Worked (stray dogs). Dog lovers pick the row (0, 15 or 30 dogs), neutrals pick the column (1, 2 or 3 walks), with payoffs (0, 5)(0, 10)(0, 15) / (10, -1)(10, 5)(10, 2) / (15, -2)(15, -4)(15, -6). Thirty dogs strictly dominates for the lovers and 1 walk is then the neutrals' best reply: the Nash outcome is **(30 dogs, 1 walk)** = (15, -2). Giving the neutrals a legal right to dog-free walks yields **(0, 15)**. Giving the lovers the right with bargaining *banned* returns (15, -2). Giving the lovers the right *with* costless bargaining maximises the total, which is 15 at both (0, 3) and **(15 dogs, 2 walks)**. The four questions together are the Coase theorem in miniature.

When Coase fails: transaction costs; many parties and free riding; asymmetric information (each side misstates its costs, so efficient deals fail); ill-defined or unenforceable rights; wealth effects. Those are exactly the cases where taxes, standards or permits are needed instead.

23 Public goods

23.1 Classification

	Excludable	Non-excludable
Rival	<i>Private</i> : food, a cinema ticket	<i>Common resource</i> : fisheries, groundwater, a congested free road
Non-rival	<i>Club</i> : cable TV, an uncongested toll road	<i>Public good</i> : national defence, a lighthouse, street lighting, basic research

Non-rivalry makes exclusion *inefficient* (charging for something that costs nothing to supply keeps people out); non-excludability makes private provision *impossible* (free riding). A common resource's failure is **over-use**; a public good's is **under-provision**. "Public good" does not mean "government-provided."

23.2 Efficient provision

Everyone consumes the same quantity, so add marginal benefits **vertically**: provide where $\sum_i MB_i(Q) = MC$ (the **Samuelson condition**). Private goods add *horizontally*. **Lindahl pricing** charges each person her own marginal benefit at Q^* ; those prices sum to MC.

Watch the kink: once a person's marginal benefit hits zero she drops out of the vertical sum, so check which branch your solution lies on.

Worked. $MB_A = 10 - Q$, $MB_B = 15 - Q$, $MC = 13$: for $Q \leq 10$, $MSB = 25 - 2Q = 13$ gives $Q^* = 6$, with Lindahl prices 4 and 9. Left to the market, only B buys, where $15 - Q = 13$, i.e. 2 units — and A free-rides. *Worked (flatmates).* $Q_a = 10 - P_a$, $Q_b = 12 - P_b$, supply perfectly elastic at 8: $MSB = 22 - 2Q = 8$ gives $Q^* = 7$, with B paying $12 - 7 = 5$ and A paying 3.

23.3 Free riding

In the voluntary-contribution game, a rupee contributed returns $m/n < 1$ to you but $m > 1$ to the group, so contributing nothing is dominant while full contribution is efficient — an n -player Prisoner's Dilemma. Experimentally, contributions start at 40–60% and **decay with repetition**; they are restored by costly peer punishment, communication, provision-point (refund) contracts, taxation, or converting the good into a club good.

Coase's diagnosis, which he has examined directly: the root causes are **undefined property rights and bargaining costs**. On that view the problem is contractual rather than technological — with well-defined rights and costless bargaining the beneficiaries would simply pay.

24 Asymmetric information: adverse selection, moral hazard, agency

The one distinction everything hangs on. **Adverse selection** is a *hidden type*: the informed party knows what she is (her health, her car's quality, her risk) *before* contracting, so the mix of people who accept a contract is worse than average. **Moral hazard** is a *hidden action*: after contracting, the informed party chooses effort, care or risk that the other side cannot observe.

If the question is about *who signs up*, it is adverse selection. If it is about *what they do afterwards*, it is moral hazard.

24.1 Adverse selection and its remedies

The lemons mechanism is in §6.5: buyers pay the expected quality of the cars *actually offered*, which depends on the price, which determines who offers, and the loop can collapse to no trade. Remedies, and what each actually fixes:

- **Mandatory pooling** — compulsory motor insurance, an individual mandate. Removes selection by removing the option to opt out.
- **Verified certification and audited disclosure** — changes beliefs. A certificate that verifies *nothing* does not, and the market still unravels.
- **Warranties and guarantees** — credible because they cost a genuinely good producer less, which is single crossing again.
- **Reputation and ratings** — turn a one-shot problem into a repeated game.
- **Screening menus** — deductibles, fare classes, plan tiers.

24.2 Moral hazard and its remedies

Full insurance destroys the incentive to take care, because the insured bears none of the loss. Standard responses all restore some link between the hidden action and the agent's own payoff: **deductibles and co-payments, experience rating, monitoring, exclusions** for specified negligence, and **performance-linked pay**. The general prescription he asks about: *tie the agent's incentives to the principal's* — which is **true** as a statement.

24.3 The classification bank

Every such question he has set, including the 2025 set, sorted.

Situation	Diagnosis
Graduates from better-branded colleges name the college on the CV; others name only the university	Adverse selection — a signal by hidden type
An institution requires 100% attendance and tracks it	Moral hazard — shirking after admission
Workplaces install CCTV to track employees	Moral hazard — monitoring a hidden action
Patients fear doctors performing needless billable procedures	Moral hazard — the doctor's incentives are not tied to outcomes
Firms hire consultants to oversee their use of AI tools	Moral hazard — a human agent carries accountability that a tool does not
Dating sites insist on current photos rather than library photos	Adverse selection — forcing accurate type revelation
Firms offer warranties on consumer durables	Adverse selection — a costly signal of quality
An ad shows a driver buying insurance seconds before a crash	Adverse selection — the keenest buyers are the worst risks
Lax airport temperature-scanning staff during COVID	Moral hazard
SEBI ties fund managers' pay to fund performance	<i>Counters</i> moral hazard
Insurers condition cover on verified medical records	<i>Fixes</i> adverse selection
Rating agencies profit from high ratings on poor securities	Moral hazard
A university publishes third-party audited placement reports	<i>A signal</i> ; reduces adverse selection
ESOPs in executive pay; incentives tied to firm goals	<i>Counter</i> moral hazard
An insured owner throws the insured necklace into the sea and claims theft	Moral hazard
Fire cover regardless of cause; repeated loan waivers; bank bailouts	<i>Create</i> moral hazard

How to write the justification in one sentence: name the hidden thing (type or action), say who cannot observe it, and say what the device does about it. For example: “Moral hazard — the employer cannot observe effort after hiring, and cameras restore observability so that shirking is deterred.”

24.4 The principal-agent problem

Setup. A risk-neutral principal cannot observe the agent's effort, only a noisy signal of it (revenue). She chooses a wage schedule contingent on that signal. Constraints: **incentive compatibility** (the agent's chosen effort is the one wanted, beating *every* alternative effort, not just the lowest), **individual rationality** (the agent's expected utility beats his outside option), and often **limited liability** (wages cannot be negative).

Three-step method. (1) Set the wage in the *bad* state to the lowest permitted value, usually zero — only the **spread** Δ provides incentives, and any level added to both states is pure cost. (2) Write the incentive constraint against *each* alternative effort and take the **most demanding** one. (3) Check participation, compute $\mathbb{E}[\text{revenue}] - \mathbb{E}[\text{wage}]$, and compare with the best “do not bother” contract.

Worked (two efforts). Revenue 100 or 20; high effort (cost 12) succeeds with probability 0.7, low effort (cost 0) with 0.3. Set $w_{20} = 0$ and $w_{100} = \Delta$. Incentive compatibility: $0.7\Delta - 12 \geq 0.3\Delta$, so $\Delta = 30$, and the agent earns a rent of 9. Profit = $0.7(100) + 0.3(20) - 0.7(30) = 55$, against 44 from accepting low effort. If the cost of high effort were 30 instead, $\Delta = 75$ and profit falls to 23.5, *below* the 44 from doing nothing — so inducing effort is not always optimal.

Worked (three efforts — the one that catches people). High / medium / low effort cost 30 / 20 / 0 and succeed with probability 0.8 / 0.5 / 0.2, revenue 100 or 0. With $\Delta = w_{100} - w_0$:

$$\text{high} = w_0 + 0.8\Delta - 30, \quad \text{medium} = w_0 + 0.5\Delta - 20, \quad \text{low} = w_0 + 0.2\Delta.$$

High beats medium iff $\Delta \geq 10/0.3 = 33.3$; high beats *low* iff $\Delta \geq 30/0.6 = 50$, which is the binding one — so the “next-best option” is low effort, not medium. With $w_0 = 0$ and $w_{100} = 50$: profit = $0.8(100) - 0.8(50) = 40$, against 20 from paying nothing. Medium effort cannot be induced at all, since it would need $\Delta \geq 66.7$ and $\Delta \leq 33.3$ simultaneously.

Worked (fixed wage versus bonus). Low effort gives revenue 10 or 60 (50–50), high effort gives 60 or 100, effort costs 10. A *fixed wage* cannot induce effort, so the agent shirks: expected revenue 35 and profit 35. Conditioning on the revenue 100, which only high effort can produce: incentive compatibility needs $\frac{1}{2}\Delta \geq 10$, so $\Delta = 20$, and profit = $80 - 10 = 70$.

Four general lessons. A risk-neutral agent with *no* limited liability can be given full incentives — effectively sell him the firm for a fixed fee, achieving the first best. With *limited liability* the principal must leave an **information rent**. With a *risk-averse* agent, incentives impose risk on someone who dislikes it, so the optimal wage is less sensitive to output the noisier the signal. Other instruments: efficiency wages, deferred pay and vesting, promotion tournaments, monitoring, and merit-based postings.

25 Templates for the verbal questions

The 2025 paper had four questions with no algebra at all, worth 40 of the 120 marks. They are the cheapest marks on the paper *if* you answer in the structure the grader is looking for.

25.1 “What positive externalities does the author describe?”

Structure each bullet as: **activity** → **who benefits** → **why the benefit is not captured by the actor**. Be specific; vague “society benefits” earns little.

The 2025 article (*federal funding of university research*) — the marked answers.

- **Life-science research** (for instance work towards eradicating malaria or curing diabetes) improves population health, and the benefit accrues to patients and to the wider economy, not to the university.
- **Basic research** — exploratory science too risky for private industry because payoffs are uncertain and decades away — has produced batteries, smartphones and the internet itself; the gains accrue to consumers and to firms that commercialise them.
- **A higher innovation return**: every \$1 million of university research yields about 1.5 patents against 0.3 for industry research, so the same money buys more innovation for the economy.
- **Knowledge spillovers to nearby firms**: businesses locate near research universities and generate more patents, which benefits those firms and the regional cluster.
- **Firm incubation**: Google, Facebook, Microsoft and Hewlett-Packard all began at universities, and the resulting products benefit consumers everywhere.

25.2 “Why is there a market failure, and why won’t private funding do?”

Structure: (i) name the failure; (ii) say why private returns fall short of social returns; (iii) say why the private sector cannot or will not fill the gap; (iv) name the remedy.

Model answer for the 2025 article, about 90 words. “Basic research is close to a public good: its output is non-rival and, once published, largely non-excludable, so the social return vastly exceeds what any single funder can appropriate. A private firm compares its own expected profit with the cost, ignoring the spillovers to other firms, to patients and to future innovation, and therefore under-invests — the standard positive-externality result. Basic research is also long-horizon and very risky, and requires funding at a scale individual firms cannot justify. Government funding internalises the spillover, which is why it is needed for efficiency.”

25.3 “Rank these pricing strategies”

Use the ranking and the two caveats in §17.5. Lead with the ranking, give one clause of justification per item, and finish with the caveat — that is what separates a 10 from a 7.

25.4 “Classify these as adverse selection or moral hazard”

Use the bank in §24.3. One sentence each: the hidden thing, who cannot observe it, and what the described device does about it.

25.5 “Write a ChatGPT prompt that would answer this”

The 2025 paper asked for prompts that would produce a brief, clear, realistic explanation of why street vendors sell *pani puri* by the plate and refuse to sell single pieces — explicitly requiring a prompt “a far distance away” from simply restating the question.

What makes such a prompt good: it supplies a *role*, the *economic frame* to use, the *constraints* of the real setting, the *form* of the answer, and a *quality bar* — without giving away the answer.

A prompt you could write:

“You are a microeconomist explaining everyday pricing to a first-year MBA class. A street food seller offers only a fixed-size plate at a single price and will not sell smaller quantities, even though splitting the portion is physically easy. Explain the demand-side and cost-side reasons a seller would design the offer this way rather than pricing per unit. Cover at least: what the per-transaction and per-customer costs look like relative to the cost of the food itself; how a fixed portion affects the seller’s ability to capture willingness to pay compared with linear pricing; what queueing and service time imply when capacity is the binding constraint; and what happens to the seller’s surplus if buyers are allowed to self-select quantity. Say what the seller is maximising. Use no jargon beyond standard terms, give one concrete numerical illustration, and keep it under 400 words.”

The economics that answer should contain (know it, in case the question asks you to evaluate the output): the real constraints are *transaction and service time*, not ingredients, so the marginal cost of a piece is tiny while the marginal cost of a *customer interaction* is large; the fixed plate is effectively a **two-part tariff collapsed into one price**, or a minimum-quantity requirement, which captures more of each buyer’s surplus than a per-piece price would; it prevents low-value buyers from consuming a single piece and occupying the same service slot as a full plate; it simplifies queue management when capacity is the binding constraint; and it standardises the product, making quality and price easy to compare, which sustains repeat custom. The seller is maximising *profit per unit of serving time*, not per piece.

26 The 2025 paper, question by question

The most recent paper and the closest guide to what is coming. Twelve questions, 10 marks each, open book, no negative marking.

Q	What it asks	Answer	Here
1	Second-price auction paid in <i>debt securities</i> ; is there a dominant strategy?	Weakly dominant to pledge $d(v) = 2v$, the bid making the winning payoff zero	\$8.1
2	The Nash equilibrium of that auction	Both bid $d(v) = 2v$ (weak dominance implies Nash)	\$8.1
3	Is it efficient?	Yes — $d(v) = 2v$ is increasing, so the highest signal wins	\$8.2
4	Debt versus cash revenue	Debt raises more: $2v/3$ against $v/2$	\$8.3
5	Would the answer change under first price?	Yes — no dominant strategy; maximise $\Pr(\text{win}) \times \text{payoff}$	\$8.4
6	Positive externalities of university research in the article	Health gains, basic research behind major technologies, 1.5 versus 0.3 patents per \$1m, spillovers to nearby firms, incubation of Google/Facebook/Microsoft	\$25.1
7	Why the market fails and why private funding is not enough	Basic research is near-public-good: non-rival, largely non-excludable, long-horizon and risky, so private returns fall far short of social returns	\$25.2
8	Rank four pricing strategies for the seller	First degree > two-part tariff > third degree > single price	\$17.5
9	Write ChatGPT prompts explaining fixed-plate pani puri pricing	Role + economic frame + real constraints + output form, without giving the answer away	\$25.5
10	Classify six situations as adverse selection or moral hazard	CV branding AS; attendance MH; CCTV MH; needless procedures MH; consultants overseeing AI MH; current photos AS	\$24.3
11a	Revenue selling OTT and insurance separately, $r_i \sim U[0, 100]$	$P = 50$ each, 25 per consumer per good $\Rightarrow$ 500,000	\$19.4
11b	Revenue from pure bundling	$P^* = \sqrt{20000/3} = 81.65, \Rightarrow$ 544,331	\$19.4
11c	Can mixed bundling beat that?	Yes — his scheme ($P_B = 81.65$, singles at 71.65) gives $\approx 546,800$; the true optimum $\approx 549,200$	\$19.4
12a	One insurance deal for two unknown utility types	Charge 500 ; both accept; profit 200	\$12.5
12b	Pay 20 for data identifying them?	Yes — tailored premiums 500 and 534.74 give profit 214.74	\$12.5

27 Answer index to the older papers

Compact, for lookup only; the full method for each is in the section named. “Q” numbers are as printed on each paper.

27.1 2024 final (descriptive)

Q	Topic	Answer
1	Pandora search over three projects	Investigate 2, then 3, then take 1; value 7.1144 (§13)
2	King of the Hill, line and ring	Dethroned iff N is odd (§3.2)
3	Permit trading, 2 units and 1 permit	$P^* = 50$ (§22.3)
4	Takeover game and worthless certification	Bid 0; both markets unravel, nobody certifies (§6.5)
5	What makes a natural monopoly	Perpetually declining AC and MC (§15.5)
6	Why price-match guarantees aid collusion	They make punishment automatic, so undercutting never pays (§5.5)
7	Write the make-or-buy profit function	$\frac{K-Q}{B}Q - c_A Q - P_c Q_2 - C(Q_1)$ (§21.2)
9	Two-part tariff vs third degree under heterogeneity	worse; better (§18.2)
10	Three true/false	True, False, False (§24, §19, §17)
11	More risk aversion: CE and risk premium	lower; higher (§12.2)

27.2 2023 final

Q	Topic	Answer
1–3	Bayesian game, expected utilities	1, $\frac{1}{2}$, 1 (§6.1)
4	Should he write an uncertain SAQ answer?	True — no negative marking, so it weakly dominates
5	Dominance in that game	Both statements false
6–10	A second pure BNE; how many	$((S, B), (S, S))$; two of them (§6.1)
11–14	Bayesian Cournot	30.46, 31.96, 32.58, 33.58 (§6.2)
15	Coase on the public-good problem	Undefined property rights and bargaining costs (§23.3)
16	Permit trading	Minimises total abatement cost without cost information (§22.2)
17–18	Doctors' incentives; verified records	Moral hazard; fixes adverse selection (§24.3)
19	Barbenheimer; mixed vs pure bundling	Both true (§19)
20	Two-part tariff claims	Both true (§18.4)
21	Stars' pay; elasticity and prices	I true, II false (§17)
22–23	Make-or-buy corner; net marginal revenue	0 engines in-house is impossible; $NMR = 6000$ (§21)
24	Joint products, total MR at $Q = 6$	1, because the negative MR truncates to zero (§20.2)
25	Rating agencies; audited placement reports	Both true (§24.3)

27.3 2022 final

Q	Topic	Answer
1–2	Core of the landowner–worker game	Not necessary: $x_2 + x_3 \geq f(2)$; sufficient: all except that (§11.2)
3–6	Core bounds	$f(n+1)$; 0; landowner with all but one worker; $f(n+1) - f(n)$
7–8	Off-by-one bound; unionisation	False; False (the landowner still needs $f(1)$)
9–13	Multi-unit auctions	Discriminatory: none of the above; Vickrey dominant: true; uniform dominant: false; $(v_i, 0)$ can be an equilibrium: true; unit demand: true (§7.4)
14	Lemon/peach trade at a pre-set price	None of the above — the answer is 0 (§6.5)
15	Externalities statements	(i) false, (ii) true (§22.1)
16	Make or buy with rising MC	None of the above — do both (§21.2)
17	Premium and perceived risk aversion	Higher (§12.3)
19–21	Third degree; two-part tariff; bundling	I true II false; both false; I true II false (§17, §18, §19)
22–24	Insurance ad; airport staff; SEBI	AS; MH; counters MH (§24.3)
25	Guessing under $+4/-1$; COVID externality	I false, II true (§12)

27.4 2021 end-term

Q	Topic	Answer
1	Warranties	Adverse selection (§24.1)
2–4	Takeover, $v \sim U[10, 30]$, $k = 1.5$	$E[v \mid \text{accept}] = 15$; profit 2.5; $B^* = 20$ (§6.5)
5–6	ESOPs; paid seat selection	Counters moral hazard; risk aversion + third degree
7–11	Hotstuff coils (transfer pricing)	1250 geysers; buy 500; sell 1500 geysers; sell 1500 coils; threshold 2500 (§21)
12–13	Dominance and efficiency statements	1 true 2 false; both false (§1)
14–17	Rose's necklace insurance	360,000; 360,000; 200,000; none of the above ($\approx 133,975$) (§12)
18	Flinging the insured necklace	Moral hazard
19–20	Public good, two flatmates	$Q^* = 7$; B pays 5 (§23.2)
21–23	Two-part tariff; bundling; discrimination	None of the above; none of the above; third degree still has DWL
24–25	One-shot price game; infinite repetition	(L, L) ; only (L, L) survives for δ near 0 (§5.4)

27.5 2020 final

Q	Topic	Answer
1–4	Shrinking-pie bargaining	A's first proposal; 520; 480; still 480 (§4.1)
5–8	Traveler's Dilemma	Maximin 2; no dominant strategy; Nash 2; one equilibrium (§2.4)
9–10	Risky investment	$E = 700,000$; maximum price = CE = 640,000 (§12.4)
11–14	Stray dogs and Coase	(30,1); (0,15); (15,-2); 15 dogs and 2 walks (§22.4)
15–18	Highlanders/Lowlanders transfer pricing	2000; none (1950); 1800; 300 (§21)
19	Two-part tariff statement	None of the above (§18.4)
20–21	Integer Bertrand	3.5; collude at 6 (§2.1)
22–24	Takeover, $v \sim U[0, 20]$, $k = 1.75$	7.5; none of the above (-1.875); bid 0 (§6.5)
25	Countering moral hazard	Tie salary incentives to firm goals

27.6 2018 and 2017 finals

Paper, Q	Topic	Answer
18-2,3	Spence signalling, three types	B.Tech; PhD (\$6.3)
18-4,5	Clinic, three effort levels	$\Delta = 50$; profit 40 (\$24.4)
18-6,7,8	11–20 game and its variant	20; none; (11, 11) (\$2.5)
18-9	Jewellery-store insurance	12.16L (\$12.4)
18-10,11	Petrol externality	$Q^* = 100$; tax 100 (\$22.1)
18-12,13	DL engines, monopoly outside market	1000; 0 (\$21.4)
18-14	Butter and buttermilk	Butter price 60 (\$20.2)
18-15	Advertising-to-sales ratio	(i) false, (ii) true (\$20.1)
18-16	Two-part tariff, two customer types	Serve the high type only; 12,800m (\$18.3)
18-17,18	Music lessons	Third degree; $ E = 1.67$ (\$17.3)
18-19,20	First degree; transfer price	All of the above; all of the above (\$17.2, \$21.1)
18-22,23	Danti and Kanti	Equal prices, positive profits; posting first raises profit (\$16.4)
18-24,25	Greasy Spoon; Lerner index	Entrée only; $ E = 5$ (\$19.2, \$15.1)
17-2	Public good	National defence (\$23.1)
17-8,9	Oil refinery joint products	$Q = 2$; $P_d = 8$ and $P_p = 32$ — label both (\$20.2)
17-10,11	Four consumers, bundling	300 with cost 40; 600 with zero cost (\$19.3)
17-12,13	Risk premium; earthquake insurance	1; 11L (\$12)
17-14,15	Owner and worker	35 with a fixed wage; 70 with a bonus (\$24.4)
17-16,17	Signalling thresholds	5; 2.5 (\$6.3)
17-18,19,20	Transfer price with a monopolised input market	6000; 2000 cars; engine price 7000 (\$21.4)
17-21,22,23	Three-firm integer Bertrand	14 equilibria; best total 7; collusion 16 (\$2.1)
17-24 to 27	Auctions with known values	FPSB: E must win, best payoff 3 (at a bid of 7); SPSB: anyone can win, best payoff 9 (\$2.2)

Two places where the printed key looks wrong, so label your work carefully if either recurs. **2017 Q9** asks for the *diesel* price and the key prints 32, which is the *petrol* price; the diesel price is 8. **2017 Q25** asks for the maximum payoff in a first-price equilibrium and the key prints 2; the correct figure is 3 (E bids 7), and the parallel 2018 quiz question, which asks for the *bid* and answers 7, confirms it.